

Hard-Constrained Structure-Preserving Learning for Inverse Diffusion: A Differentiable Galerkin Route to Option Calibration

Abstract

Scientific machine learning for inverse partial differential equations faces an architectural choice: physical admissibility may be encouraged by a loss, approximated by a black-box solution operator, or enforced by the numerical solver itself. We take the third route for inverse diffusion, motivated by local-volatility calibration. A neural field parameterises only a strictly positive diffusion coefficient, while option-like prices are produced by a differentiable Galerkin time stepper. Positivity, coercivity, symmetric positive-definiteness (SPD) of every implicit layer, and feasibility of the associated obstacle solve are therefore properties of the architecture rather than penalties that can be traded against data fit. The resulting coefficient-to-solution map, called the σ -Q operator, is differentiated through its SPD solves and paired with explicit statistical regularisation of the inverse problem. Exact weighted-Dupire layers support coefficient recovery, flux-density and nested-sample studies, including maturity and contiguous-strike holdouts on an SPX snapshot. Separate symmetric divergence-form benchmarks isolate noise regularisation and the strictly convex American-style obstacle problem. Together the studies assess coefficient recovery, mesh refinement, noise sensitivity, deployable regularisation, sample-size scaling, cleaned early-exercise geometry and out-of-sample market calibration.

We then probe the architectural thesis in four additional studies. A mechanism-level comparison with PINN, Deep Ritz and VPINN formulations is consistent with a no-go proposition we prove here—no finite penalty weight can certify admissibility. The hard and soft branches deliberately share the coefficient-network family, data and optimiser, but they do not form a fully controlled architecture comparison: the soft methods also learn an independent state network and use formulation-specific objectives. Across the five-seed aggregate the soft coefficients become negative despite small own-state price errors, while all reported formal runs remain factorable; a separate seed-11 layer-profile diagnostic shows that one PINN fit can nevertheless produce indefinite implicit layers and failed factorisations. A zero-shot study against FNO, DeepONet and a graph neural operator gives a deliberately mixed verdict: the learned operators are more accurate in distribution, because they learn to correct discretisation error, but degrade under horizon extrapolation while emitting negative and non-monotone prices. One post-hoc σ -Q SPD correction removes the tested box violations but does not certify the full no-arbitrage set or restore monotonicity. A dual-density study proves and then verifies that the Dupire dual is exactly the Fokker–Planck equation of a log-forward diffusion, and shows that differentiating the solver yields a training-free score model competitive with score-matching and DDPM baselines on the declared benchmark. A Gauss–Newton–Laplace study delivers credible bands for the coefficient and the Arrow–Debreu density, validated against a parametric bootstrap and two MCMC samplers, with a sandwich-covariance identity that explains analytically why the bootstrap spread must fall below the posterior spread.

Beyond the application, the construction identifies a reusable SciML design principle: learn the unknown coefficient inside a structure-preserving solver rather than learn around the solver. We keep the remaining scope explicit. E1, E2, E6 and E7 use the exact weighted Dupire layer, whereas E3–E5 remain controlled symmetric divergence-form benchmarks and E8–E11 retain their stated study-specific operators. Entropic martingale transport and rough or nonlocal

extensions are derived as interfaces rather than evaluated. This separation between proved results, measured results and testable extensions turns an inverse-diffusion solver into a broader research programme in hard-constrained neural PDE methods.

1 Introduction

Scientific machine learning for inverse PDE problems faces a basic design choice: structure can be encouraged through a loss, represented by a learned solution operator, or enforced by the numerical architecture. The distinction is consequential. A small residual penalty does not imply that a trial coefficient is admissible; a highly accurate black-box surrogate need not preserve coercivity; and neither property alone guarantees that gradients of the inverse solve remain meaningful. This paper develops the third alternative: *learn only the unknown coefficient and leave evolution to a differentiable, structure-preserving Galerkin solver.*

Local-volatility calibration is a demanding test bed for this idea. Given an arbitrage-consistent call surface $C(T, K)$, the Dupire programme seeks a deterministic diffusion coefficient $\sigma_{\text{loc}}(T, K)$ that reproduces European vanilla prices [1, 2]. In market data, however, quotes are noisy, irregular and confined to a sparse lattice. Direct Dupire inversion requires strike and maturity derivatives and is therefore unstable. Industry practice first fits a smooth implied-volatility surface, often with SVI or SSVI [4, 5], and differentiates the fit afterwards. That pipeline is effective as interpolation but separates the inferred coefficient from the numerical operator subsequently used for pricing and risk.

Machine-learning methods increase flexibility but place structure at different levels. PINNs impose strong- or weak-form residuals as soft objectives [7, 8]; Deep Ritz methods optimise a variational energy [9]; DeepONet and FNO learn maps between function spaces [11, 12]. Structure-preserving networks instead encode a Hamiltonian, Lagrangian, or port-Hamiltonian representation in the model class [13, 14, 15]. Our contribution belongs to this last family but targets a dissipative inverse PDE: admissibility is encoded in the coefficient parameterisation, and the state is obtained from an implicit Galerkin layer whose matrix is SPD by construction.

Architectural thesis. Let f_θ be a neural or nodal field and set $\kappa_\theta = \text{softplus}(f_\theta) + \kappa_{\min}$. The learned object is κ_θ ; the price is not another network output but the solution of a finite-element evolution assembled from that field. On the symmetric coercive formulation studied below, this separation yields three levels of constraint. Coefficient positivity is *hard*; the discrete PDE is satisfied to linear-solver and discretisation tolerance; data fit and smoothness remain *statistical* objectives. Conflating these levels obscures what a SciML method actually guarantees.

Contributions.

1. **A hard-constrained coefficient-to-solution architecture.** Strictly positive interface coefficients give a coercive weak form and an SPD implicit layer. These properties hold for every parameter value and are not loss-dependent.
2. **A no-go proposition for soft physics penalties.** We prove that for any continuous penalty vanishing exactly on the admissible set and any finite weight, the sublevel sets of the penalised objective always contain inadmissible coefficients (3), whereas the positivity reparameterisation makes the inadmissible set empty. Soft and hard constraints are therefore not two points on a spectrum.

3. **Quantitative discrete structure, including its limits.** We give the exact layer spectrum and condition-number bound (4) and prove that layer SPD is *strictly weaker* than coefficient positivity (1): a lumped mass can mask an inadmissible coefficient, which is why we audit both quantities.
4. **A differentiable Galerkin inverse solver with a complete proof.** The σ -Q map propagates calibration gradients through the same sparse SPD solves used in the forward pass; 5 is proved by the implicit function theorem on the block lower-bidiagonal state Jacobian, together with the adjoint recursion, rather than by identifying invertibility with good conditioning.
5. **The dual density and its exact diffusion interpretation.** 6 proves that the Dupire dual is the density of the log-forward return in divergence form, has unit mass, and satisfies exactly the Fokker–Planck equation of a log-forward SDE; 7 derives its Anderson reverse-time drift, which contains an extra transport term absent from constant-variance schedules.
6. **A shared variational backbone.** The assembled operator supports European propagation, a grid-based density recovery diagnostic, and a strictly convex American obstacle solve. We state positivity and density claims as diagnostics unless a separate monotonicity or convexity proof is available.
7. **Four controlled studies of the thesis itself.** E8 (10.7) compares hard against PINN, Deep Ritz and VPINN under matched data, optimiser and budget; E9 (10.8) measures zero-shot generalisation against FNO, DeepONet and a graph neural operator across four distribution-shift regimes; E10 (10.9) verifies the dual-density propositions and benchmarks a training-free score model against score matching and DDPM; E11 (10.10) validates Gauss–Newton–Laplace credible bands against a parametric bootstrap and two MCMC samplers, with a sandwich-covariance identity (8) that predicts the observed bootstrap/posterior ratio.
8. **Explicit residual scope.** E1, E2, E6 and E7 use the exact weighted Dupire operator; E3–E5 isolate the symmetric divergence-form mechanism, and E8–E11 use the operators stated in their individual protocols. The entropic martingale bridge and rough or nonlocal extensions are derived as interfaces in C rather than evaluated. We report the regimes in which learned operators beat the solver as prominently as those in which they fail.

The broader message is that neural flexibility need not replace numerical structure. For inverse PDE learning it can be confined to the unknown coefficient while admissibility, evolution and differentiation remain governed by a variational solver.

2 Related work

Local-volatility inversion. Dupire and Derman–Kani introduced the local-volatility representation [1, 2]; regularised coefficient recovery addresses its ill-posedness [6]. SVI/SSVI provide effective arbitrage-aware surface parameterisations [4, 5]. Our distinction is architectural: coefficient regularisation is coupled to a differentiable variational solver.

Variational and structure-preserving SciML. PINNs, hp-VPINNs and Deep Ritz place differential or variational information in training objectives [7, 8, 9]. Hamiltonian, Lagrangian and port-Hamiltonian networks hard-code dynamical structure [13, 14, 15]. Classical FEM theory already explains why a positive symmetric elliptic coefficient yields SPD; PDE-constrained optimisation already supplies parameter-to-state differentiation [17, 18]. The claimed novelty is therefore

not SPD or implicit differentiation separately, but their combination in a positivity-parameterised inverse-diffusion architecture that reuses one variational object across calibration and obstacle solves. What we add to this literature is a proof that the soft and hard routes are qualitatively rather than quantitatively different (3), a quantitative statement of how far discrete SPD does *not* certify admissibility (1), and a mechanism-level empirical comparison with three soft formulations (10.7).

Neural operators. DeepONet, FNO and the graph kernel operator family motivate amortisation across problem instances, and they formalise the notion of discretisation invariance [11, 12, 33]. 10.8 contributes a paired-seed comparison under shift in which the learned operators are more accurate in distribution and structurally unsafe outside it, and shows that one post-hoc σ -Q/SPD correction reduces targeted violations without restoring either accuracy or sampled-grid monotonicity.

Score-based diffusion. Anderson’s reverse-time theorem underlies score-based SDEs and denoising diffusion probabilistic models [20, 21, 30]; implicit and denoising score matching are the two standard estimators of the score [31, 32]. Our contribution is to show that the Dupire dual supplies the same forward Fokker–Planck object exactly (6), with a state-dependent diffusion whose reversal carries an extra transport term (7), and that differentiating a structure-preserving solve yields a training-free score competitive with both estimators and with a tuned DDPM on a tractable benchmark (10.9).

Bayesian deep learning and UQ for scientific ML. Bayesian inverse problems in function space, and their sampling algorithms, provide the reference framework [34, 35, 36]. Bayesian PINNs, Laplace approximations for deep networks and SciML-UQ surveys provide the modern comparators [27, 29, 28]. 10.10 follows the standard validation protocol for such approximations, namely a parametric bootstrap [37] and MCMC with split- $\hat{R}$ and effective-sample-size diagnostics [38, 39], and adds a sandwich-covariance identity (8) in the spirit of misspecification-robust asymptotics [40] that predicts the bootstrap/posterior spread ratio rather than merely observing it. MOT and Schrödinger bridges remain motivating rather than evaluated [22, 23].

3 Risk-neutral setup, weighted symmetry, and modelling scope

Let S_t follow the risk-neutral local-volatility diffusion

$$dS_t = (r - q)S_t dt + \sigma_{\text{loc}}(t, S_t)S_t dW_t. \quad (1)$$

Let $D(0, T)$ and F_T denote the discount factor and forward, and define the normalised forward call $c(T, k) = C(T, F_T e^k) / (D(0, T)F_T)$ with $k = \log(K/F_T)$. For deterministic rates and dividends, the exact log-moneyness Dupire equation is

$$\partial_{TC} = \kappa(T, k)(\partial_{kk}c - \partial_k c), \quad \kappa = \frac{1}{2}\sigma_{\text{loc}}^2, \quad (2)$$

with $c(0, k) = (1 - e^k)_+$. The first derivative in (2) is essential: a numéraire change alone does not turn Dupire into $\partial_k(\kappa\partial_k c)$.

For fixed T , however, the operator is symmetric in a coefficient-dependent weighted space. Set

$$m_T(k) = \frac{e^{-k}}{\kappa(T, k)}, \quad w(k) = e^{-k}. \quad (3)$$

Multiplying (2) by m_T gives

$$m_T \partial_T c = \partial_k (w \partial_k c). \quad (4)$$

After subtracting a Dirichlet lift, the spatial bilinear form $a(c, v) = \int_{\Omega} w c_k v_k dk$ is symmetric and coercive. A P_1 backward-Euler discretisation therefore has the SPD layer

$$(\mathbf{M}(m_T^n) + \Delta T_n \mathbf{K}(w)) c^n = \mathbf{M}(m_T^n) c^{n-1} + b^n, \quad (5)$$

provided $0 < \kappa_{\min} \leq \kappa \leq \kappa_{\max} < \infty$. Here the learned coefficient enters the positive weighted mass matrix, while the stiffness uses $w = e^{-k}$. This is the mathematically exact structure-preserving Dupire layer. The lower ellipticity bound is enforced by softplus; the upper bound is an additional analytical or architectural assumption.

Experimental operators. The experiments deliberately use two related operators. E1, E2, E6 and E7 use the exact weighted layer (5), including the financial density $q = e^{-k}(c_{kk} - c_k)$. E3–E5 retain the symmetric coefficient-diffusion benchmark

$$\partial_{\tau} u = \partial_k (\kappa(\tau, k) \partial_k u), \quad (6)$$

generated by

$$a_{\kappa}(u, v) = \int_{\Omega} \kappa u_k v_k dk, \quad \Omega = (k_{\min}, k_{\max}). \quad (7)$$

The latter isolates the paper’s central SciML mechanism—a learned positive coefficient assembled into an SPD Galerkin operator—but is not presented as an exact local-volatility equation. Every numerical subsection below names its operator, and no result is transferred silently between the two formulations.

Proposition 1 (Parabolic well-posedness). *Suppose $\kappa \in L^{\infty}((0, T) \times \Omega)$ and $0 < \kappa_{\min} \leq \kappa \leq \kappa_{\max} < \infty$. After a sufficiently regular boundary lift, both (4) and (6) define non-autonomous continuous coercive forms. Standard parabolic evolution theory then gives a unique weak solution in $L^2(0, T; H_0^1(\Omega)) \cap H^1(0, T; H^{-1}(\Omega))$ with continuous dependence on the data.*

4 P_1 finite elements and the learnable stiffness matrix

Partition Ω into $N - 1$ uniform cells of width h with nodes $k_0 < \dots < k_{N-1}$ and hat basis $\{\phi_i\}$. A conforming P_1 (Galerkin) discretisation of (7) produces the tridiagonal *stiffness matrix*

$$\mathbf{K}(\kappa)_{ii} = \frac{\kappa_{i-\frac{1}{2}} + \kappa_{i+\frac{1}{2}}}{h}, \quad \mathbf{K}(\kappa)_{i,i\pm 1} = -\frac{\kappa_{i\pm\frac{1}{2}}}{h}, \quad (8)$$

where the interface values use the arithmetic mean $\kappa_{i+\frac{1}{2}} = \frac{1}{2}(\kappa_i + \kappa_{i+1})$. The arithmetic mean is a deliberate choice: it makes $\mathbf{K}$ linear in the nodal values κ_i , so $\partial \mathbf{K} / \partial \kappa$ is constant and the operator is exactly (not just approximately) differentiable. The mass matrix is lumped, $\mathbf{M} = \text{diag}(h)$, which keeps time-stepping monotone (an M -matrix).

Proposition 2 (Hard sufficient condition for SPD). *If every interface coefficient satisfies $\kappa_{i+\frac{1}{2}} \geq \kappa_{\min} > 0$, then the interior stiffness matrix $\mathbf{K}(\kappa) \in \mathbb{R}^{(N-2) \times (N-2)}$ from (8) is symmetric positive-definite.*

Proof. Symmetry is immediate. For any nonzero interior vector v extended by the homogeneous boundary values,

$$v^\top \mathbf{K} v = \sum_i \frac{\kappa_{i+\frac{1}{2}}}{h} (v_{i+1} - v_i)^2 \geq \kappa_{\min} \sum_i \frac{(v_{i+1} - v_i)^2}{h} > 0.$$

Thus positive interface weights are a transparent architectural condition for SPD. They are sufficient rather than necessary: a global matrix can remain SPD for some sign-indefinite weights, but that larger spectral admissible set is not used here. $\square$

2 converts the analytic lower-ellipticity requirement into an algebraic guarantee for every optimisation iterate. It guarantees uniqueness of the European-style state layer, invertibility of the state-equation Jacobian, and strict convexity of the corresponding obstacle QP. It does not imply that the coefficient-to-observation map is invertible, nor does it guarantee a mesh-independent condition number. We enforce the sufficient condition structurally:

$$\kappa_\theta(\tau, k) = \text{softplus}(f_\theta(\tau, k)) + \kappa_{\min}, \quad \kappa_{\min} = \frac{1}{2} \sigma_{\min}^2, \quad (9)$$

with $\sigma_{\min} = 3\%$. Here f_θ is either a small multilayer perceptron or a free nodal tensor. An upper ellipticity bound, when required for uniform estimates, can be imposed with a scaled sigmoid or treated as an explicit analytical assumption.

The following elementary but load-bearing statement is what we mean by “hard” as opposed to “soft”. It is the formal content of the comparison in C.1 and the property that E8 measures.

Proposition 3 (Soft penalties cannot certify admissibility). *Let Θ be a parameter space, let $\theta \mapsto \kappa_\theta$ be continuous into the interface values $(\kappa_{i+\frac{1}{2}})_i$, and let*

$$\mathcal{A} = \{\theta : \kappa_{i+\frac{1}{2}}(\theta) \geq \kappa_{\min} \ \forall i\}$$

be the admissible set. Let $P : \Theta \rightarrow [0, \infty)$ be any continuous penalty with $P^{-1}(0) = \mathcal{A}$, for instance $P(\theta) = \sum_i (\kappa_{\min} - \kappa_{i+\frac{1}{2}}(\theta))_+^p$ with $p \geq 1$, and let $L_\lambda = L_{\text{data}} + \lambda P$ with L_{data} continuous. Assume $\partial \mathcal{A} \neq \emptyset$ and that κ_θ is locally surjective near some $\theta_0 \in \partial \mathcal{A}$. Then for every finite λ and every $\varepsilon > 0$ the sublevel set $\{\theta : L_\lambda(\theta) \leq L_\lambda(\theta_0) + \varepsilon\}$ contains parameters θ with $\min_i \kappa_{i+\frac{1}{2}}(\theta) < 0$. By contrast, under (9) one has $\Theta = \mathcal{A}$, so the inadmissible set is empty for every θ , every loss and every training budget.

Proof. Since $\theta_0 \in \partial \mathcal{A}$ we have $P(\theta_0) = 0$. By local surjectivity choose a sequence $\theta_t \rightarrow \theta_0$ with $\min_i \kappa_{i+\frac{1}{2}}(\theta_t) = -\delta_t < 0$ and $\delta_t \downarrow 0$. Then $0 \leq P(\theta_t) \leq C\delta_t^p \rightarrow 0$ and, by continuity, $L_{\text{data}}(\theta_t) \rightarrow L_{\text{data}}(\theta_0)$; hence $L_\lambda(\theta_t) \rightarrow L_\lambda(\theta_0)$ for each fixed λ . So all but finitely many θ_t lie in the stated sublevel set while violating positivity. For the second claim, $\text{softplus} > 0$ gives $\kappa_\theta \geq \kappa_{\min}$ pointwise, hence the same bound for the arithmetic interface means, for every $\theta \in \Theta$. $\square$

Remark 1. 3 is not an argument that penalties are useless; it identifies what they can and cannot deliver. A penalty makes violation *expensive* and therefore rare at a good optimum, but admissibility remains a property of the achieved iterate, verifiable only after the fact. Consequently a soft-constrained method must be audited by measuring $\min_i \kappa_{i+\frac{1}{2}}$, the per-layer spectra and the factorisation success rate — exactly the diagnostics reported in 10.7 — rather than by reporting the penalty value, which is small in both the admissible and the marginally inadmissible regime.

5 The σ -Q operator

We march (6) by backward Euler. With interior unknowns u^n at maturity τ_n and step $\Delta\tau_n = \tau_n - \tau_{n-1}$,

$$(\mathbf{M} + \Delta\tau_n \mathbf{K}(\kappa^n)) u^n = \mathbf{M} u^{n-1} + (\text{Dirichlet lift}). \quad (10)$$

The system matrix $\mathbf{A}_n = \mathbf{M} + \Delta\tau_n \mathbf{K}$ is SPD (2), tridiagonal, and solved in $O(N)$ by the Thomas algorithm.

Proposition 4 (Quantitative layer spectrum). *Let $\mathbf{M} = h\mathbf{I}$ be the lumped interior mass matrix and $\mathbf{L} = \text{tridiag}(-1, 2, -1) \in \mathbb{R}^{(N-2) \times (N-2)}$. If $\kappa_{\min} \leq \kappa_{i+\frac{1}{2}} \leq \kappa_{\max}$ for all i , then*

$$h + \frac{\Delta\tau_n \kappa_{\min}}{h} \lambda_{\min}(\mathbf{L}) \leq \lambda_{\min}(\mathbf{A}_n), \quad \lambda_{\max}(\mathbf{A}_n) \leq h + \frac{4\Delta\tau_n \kappa_{\max}}{h}, \quad (11)$$

with $\lambda_{\min}(\mathbf{L}) = 4 \sin^2(\pi/(2N-2))$, and consequently

$$\text{cond}(\mathbf{A}_n) \leq 1 + \frac{4\Delta\tau_n \kappa_{\max}}{h^2}. \quad (12)$$

In particular $\lambda_{\min}(\mathbf{A}_n) \geq h > 0$ uniformly in the coefficient, and $\mathbf{A}_n$ is a symmetric M-matrix, so the layer solve is monotone.

Proof. Both bounds follow from the Rayleigh quotient. The proof of 2 gives the quadratic form of the stiffness matrix, hence

$$v^\top \mathbf{K} v = \sum_i h^{-1} \kappa_{i+\frac{1}{2}} (v_{i+1} - v_i)^2, \quad \kappa_{\min} h^{-1} v^\top \mathbf{L} v \leq v^\top \mathbf{K} v \leq \kappa_{\max} h^{-1} v^\top \mathbf{L} v,$$

because $v^\top \mathbf{L} v = \sum_i (v_{i+1} - v_i)^2$ with the homogeneous extension. The extreme eigenvalues of $\mathbf{L}$ are $4 \sin^2(\pi/(2N-2))$ and $4 \cos^2(\pi/(2N-2)) < 4$. Adding $\mathbf{M} = h\mathbf{I}$ shifts the whole spectrum by h , which yields (11); (12) follows by dividing the two bounds and using $\lambda_{\min}(\mathbf{A}_n) \geq h$. The off-diagonal entries of $\mathbf{A}_n$ equal $-\Delta\tau_n \kappa_{i\pm\frac{1}{2}}/h$, which is ≤ 0 , and $\mathbf{A}_n$ is SPD, hence a symmetric M-matrix with $\mathbf{A}_n^{-1} \geq 0$ entrywise. $\square$

Corollary 1 (Layer SPD is strictly weaker than positivity). *For a given coefficient vector, $\mathbf{A}_n = h\mathbf{I} + \Delta\tau_n \mathbf{K}$ is SPD if and only if $\lambda_{\min}(\mathbf{K}) > -h/\Delta\tau_n$. Hence a method that produces interface values with $\min_i \kappa_{i+\frac{1}{2}} < 0$ can still yield SPD layers whenever the negativity is small relative to $h/\Delta\tau_n$, and can lose SPD only where it is not.*

Proof. Immediate from $\lambda_{\min}(\mathbf{A}_n) = h + \Delta\tau_n \lambda_{\min}(\mathbf{K})$ and the fact that $\mathbf{K}$ is symmetric. $\square$

1 matters for how the hard-constraint claim must be stated and tested. Coefficient positivity implies layer SPD, but not conversely: the lumped mass regularises the layer and can mask an inadmissible coefficient. The architectural guarantee therefore lives at the level of the coefficient, and a soft-constrained competitor may pass a naive SPD check on some layers while its coefficient is physically meaningless. E8 accordingly reports both quantities, and finds both failure modes empirically (10.7).

The map

$$\mathcal{Q}: \kappa \mapsto (u^0, u^1, \dots, u^{N_\tau}) \quad (13)$$

is the σ -Q operator: a differentiable discrete pricing map from a diffusion field to a full price surface.

Proposition 5 (Discrete Fréchet differentiability and adjoint). *Let $\mathcal{U}$ be an open coefficient set on which every layer matrix is SPD. If $\mathbf{A}_n(\kappa)$ and its right-hand side $b_n(\kappa, u^{n-1})$ are continuously differentiable on $\mathcal{U}$, then the full finite-dimensional time-stepping map is continuously Fréchet differentiable. For a perturbation $\delta\kappa$, its layerwise directional derivative satisfies*

$$Du^n[\delta\kappa] = \mathbf{A}_n^{-1}(Db_n[\delta\kappa] - D\mathbf{A}_n[\delta\kappa]u^n), \quad (14)$$

where Db_n includes dependence through u^{n-1} and the Dirichlet lift. For any smooth scalar loss, a reverse sweep reuses transposed layer solves; these equal the forward SPD systems in the symmetric benchmark.

Proof. Write the stacked state $U = (u^1, \dots, u^{N_\tau})$ and define $F : \mathcal{U} \times \mathbb{R}^d \rightarrow \mathbb{R}^d$, $d = (N-2)N_\tau$, layerwise by

$$F_n(\kappa, U) = \mathbf{A}_n(\kappa)u^n - b_n(\kappa, u^{n-1}), \quad n = 1, \dots, N_\tau,$$

with u^0 the prescribed initial datum. Each $\mathbf{A}_n$ is affine in the nodal coefficient values, because the arithmetic interface mean makes (8) linear in κ_i , and b_n is affine in u^{n-1} and in the lift; hence F is C^∞ in (κ, U) and in particular C^1 on $\mathcal{U} \times \mathbb{R}^d$. The partial Jacobian $\partial F / \partial U$ is block lower bidiagonal with diagonal blocks $\mathbf{A}_n$ and subdiagonal blocks $-\partial b_n / \partial u^{n-1} = -\mathbf{M}$. A block triangular matrix is invertible whenever its diagonal blocks are, and each $\mathbf{A}_n$ is SPD on $\mathcal{U}$ by 2; therefore $\partial F / \partial U$ is invertible on $\mathcal{U}$. The implicit function theorem applied to $F(\kappa, U) = 0$ gives a unique C^1 solution map $\kappa \mapsto U(\kappa)$ on $\mathcal{U}$ with

$$DU[\delta\kappa] = -\left(\frac{\partial F}{\partial U}\right)^{-1} \frac{\partial F}{\partial \kappa}[\delta\kappa].$$

Because $\partial F / \partial U$ is block triangular, forward substitution turns this identity into the layerwise recursion (14), with $Db_n[\delta\kappa] = \mathbf{M} Du^{n-1}[\delta\kappa] + D(\text{lift})[\delta\kappa]$. Continuity of DU follows from continuity of $\kappa \mapsto \mathbf{A}_n^{-1}$ on the SPD set. For a scalar loss $\ell(U)$, transposing the recursion gives the adjoint sweep

$$\mathbf{A}_n^\top p^n = \partial_{u^n} \ell + \mathbf{M}^\top p^{n+1}, \quad p^{N_\tau+1} = 0, \quad \nabla_\kappa \ell = - \sum_n (D\mathbf{A}_n[\cdot] u^n - Db_n[\cdot])^\top p^n,$$

which runs backwards in n and reuses the same factorisations since $\mathbf{A}_n^\top = \mathbf{A}_n$. Each $D\mathbf{A}_n[\cdot]$ is a constant sparse tensor by linearity of (8), so the derivative is exact rather than a finite-difference surrogate. $\square$

Automatic differentiation through the tridiagonal solve implements (14). SPD guarantees invertibility and permits stable Cholesky- or Thomas-type algorithms, while quantitative gradient conditioning additionally depends on the mesh and the ellipticity ratio. The proposition concerns the discrete solver; continuous coefficient-to-state differentiability requires the corresponding function-space assumptions from PDE-constrained optimisation [18].

6 Losses: data fit, energy consistency, smoothness

Data fit. Market quotes are implied vols $\hat{\sigma}_j$ at nodes (τ_{a_j}, k_{b_j}) . We map them once to model-consistent prices p_j^* through a monotone price-vol table built *from the operator itself* (so the target lives in the operator's own price space), and penalise $L_{\text{data}} = \sum_j \omega_j (u_{a_j, b_j} - p_j^*)^2$ with vega weights ω_j that down-weight illiquid deep wings.

Solver-energy diagnostic. For the symmetric benchmark (6), homogeneous lifted boundary data give

$$\frac{1}{2} \frac{d}{d\tau} \|u(\tau)\|_{L^2}^2 + u^\top \mathbf{K}(\kappa) u = 0. \quad (15)$$

Backward Euler satisfies the sharper discrete identity

$$\frac{\|u^n\|_{\mathbf{M}}^2 - \|u^{n-1}\|_{\mathbf{M}}^2 + \|u^n - u^{n-1}\|_{\mathbf{M}}^2}{2\Delta\tau_n} + (u^n)^\top \mathbf{K}(\kappa^n) u^n = 0, \quad (16)$$

up to boundary work and solver tolerance. We therefore use the residual of (16) as a numerical consistency diagnostic, not as a substitute for solving the PDE. For the exact weighted Dupire layer (5), the time dependence of m_T contributes an additional term $\frac{1}{2} \int (\partial_T m_T) c^2 dk$.

Coefficient regularisation. The actual regulariser of the ill-posed coefficient inversion is

$$L_{\text{smooth}} = \|\partial_{kk}\kappa\|_{L^2}^2 + \|\partial_{\tau\tau}\kappa\|_{L^2}^2,$$

together with any prior or box constraint on the coefficient. Writing L_{disc} for the squared residual of (16), the composite objective is

$$L = (1 - \lambda)L_{\text{data}} + \lambda L_{\text{disc}} + \mu L_{\text{smooth}}. \quad (17)$$

Remark 2 (Hard constraints versus regularisation). The positivity parameterisation and SPD solve are hard architectural properties. The discrete residual is a verification signal and is nearly self-satisfied when the layer solve is accurate. The coefficient penalty with weight μ controls the bias–variance trade-off of the inverse problem; E3–E4 assess that distinct statistical role.

7 Dual density as a post-processing diagnostic

Breeden–Litzenberger gives $\partial_{KK}C(T, K) = D(0, T)f_{S_T}(K)$ in strike coordinates [3]. For the normalised forward call $c(T, k)$ of 3 the corresponding object is the density of the log-forward return, and it satisfies a conservative forward equation. Both statements are elementary but they are the exact bridge to C.3, so we record them with proof.

Proposition 6 (Dual density, flux form, and the forward equation). *Let $c(T, \cdot) \in C^2$ solve the exact log-moneyness Dupire equation (2) with $c(T, k) \rightarrow 0$ and $\partial_k c(T, k) \rightarrow 0$ as $k \rightarrow +\infty$, and $c(T, k) - (1 - e^k) \rightarrow 0$ as $k \rightarrow -\infty$. Define*

$$q_T(k) = e^{-k} (\partial_{kk}c(T, k) - \partial_k c(T, k)). \quad (18)$$

Then

- (i) q_T is the density of $X_T = \log(S_T/F_T)$ under the T -forward measure, and it admits the divergence form $q_T = \partial_k(e^{-k}\partial_k c)$;
- (ii) $\int_{\mathbb{R}} q_T(k) dk = 1$ for every T ;
- (iii) q solves the conservative forward equation

$$\partial_T q = \partial_k(\kappa q) + \partial_{kk}(\kappa q), \quad (19)$$

which is exactly the Fokker–Planck equation of the log-forward diffusion

$$dX_T = -\kappa(T, X_T) dT + \sqrt{2\kappa(T, X_T)} dW_T. \quad (20)$$

Proof. (i) With $K = F_T e^k$ and $c = C/(DF_T)$ one has $\partial_K = F_T^{-1} e^{-k} \partial_k$, hence

$$\partial_{KK} C = D(0, T) F_T^{-1} e^{-2k} (\partial_{kk} c - \partial_k c).$$

Since the density of X_T is $f_{S_T}(K) dK/dk = F_T e^k f_{S_T}(K)$, combining with Breeden–Litzenberger gives precisely (18). The divergence form follows from $\partial_k(e^{-k} \partial_k c) = e^{-k} (\partial_{kk} c - \partial_k c)$.

(ii) By (i), $\int_{-R}^R q_T = [e^{-k} \partial_k c]_{-R}^R$. As $k \rightarrow +\infty$, $\partial_k c \rightarrow 0$ faster than e^k grows for a call with finite price, so the upper term vanishes; as $k \rightarrow -\infty$, $c \rightarrow 1 - e^k$ gives $\partial_k c \rightarrow -e^k$ and $e^{-k} \partial_k c \rightarrow -1$. Hence the bracket tends to $0 - (-1) = 1$.

(iii) Differentiate (18) in T and substitute (2). Writing $\mathcal{L}c = \partial_{kk} c - \partial_k c$ and noting that $\mathcal{L}$ commutes with ∂_T ,

$$\partial_T q = e^{-k} \mathcal{L}(\partial_T c) = e^{-k} \mathcal{L}(\kappa \mathcal{L}c) = e^{-k} \mathcal{L}(e^k \kappa q).$$

Put $g = \kappa q$. Then $e^{-k} \mathcal{L}(e^k g) = e^{-k} (\partial_{kk}(e^k g) - \partial_k(e^k g)) = e^{-k} (e^k g + 2e^k \partial_k g + e^k \partial_{kk} g - e^k g - e^k \partial_k g) = \partial_k g + \partial_{kk} g$, which is (19). Finally, the Fokker–Planck equation of $dX = b dT + \varsigma dW$ is $\partial_T q = -\partial_k(bq) + \frac{1}{2} \partial_{kk}(\varsigma^2 q)$; matching $b = -\kappa$ and $\varsigma^2 = 2\kappa$ gives (20). The drift $-\kappa = -\frac{1}{2} \sigma_{\text{loc}}^2$ is the Itô correction that keeps $\mathbb{E}[e^{X_T}] = 1$, i.e. the forward-martingale property. $\square$

Equations (2) and (19) are the backward pricing/forward-density dual pair that motivates the diffusion-model connection in C.3, and 6(ii) supplies a mass-conservation invariant that we use as a hard structural check in E10.

The forward pair also determines the reverse-time dynamics exactly, which is the object a score-based generative model has to learn. We record it because the state-dependent diffusion coefficient produces two score terms rather than the single term familiar from constant-variance schedules.

Proposition 7 (Reverse-time dynamics of the calibrated diffusion). *Fix a horizon $\bar{T}$ and let q be the positive $C^{1,2}$ solution of (19) on $(0, \bar{T}] \times \mathbb{R}$ with $\kappa \in C^1$ and $\kappa \geq \kappa_{\min} > 0$. Let $\bar{W}$ be a Brownian motion in the reversed time variable. Then the time reversal $\bar{X}_s = X_{\bar{T}-s}$ of (20) solves*

$$d\bar{X}_s = \mu_{\text{rev}}(\bar{T} - s, \bar{X}_s) ds + \sqrt{2\kappa(\bar{T} - s, \bar{X}_s)} d\bar{W}_s, \quad (21)$$

with

$$\mu_{\text{rev}}(T, k) = -\left(-\kappa(T, k) - \partial_k(2\kappa(T, k)) - 2\kappa(T, k) \partial_k \log q_T(k)\right) = \kappa + 2\partial_k \kappa + 2\kappa \partial_k \log q_T. \quad (22)$$

Proof. Anderson’s reverse-time theorem [20] states that for $dX = b dT + \varsigma dW$ with density q , the reversed process is a diffusion with the same diffusion coefficient ς and forward-time drift

$$\tilde{b}(T, k) = b(T, k) - \frac{1}{q} \partial_k(\varsigma^2 q)(T, k),$$

the reversal being obtained by running $\tilde{b}$ backwards. With $b = -\kappa$ and $\varsigma^2 = 2\kappa$ from 6(iii),

$$\frac{1}{q} \partial_k(2\kappa q) = 2\partial_k \kappa + 2\kappa \frac{\partial_k q}{q} = 2\partial_k \kappa + 2\kappa \partial_k \log q,$$

so $\tilde{b} = -\kappa - 2\partial_k \kappa - 2\kappa \partial_k \log q$. Negating for the reversed time variable gives (22); positivity of κ and of q makes every term well defined, and $\kappa \geq \kappa_{\min} > 0$ guarantees the reversed diffusion is uniformly nondegenerate. $\square$

Remark 3 (Two score terms and a training-free estimator). For a constant schedule $\partial_k \kappa = 0$, so (22) reduces to the familiar $\kappa + 2\kappa \partial_k \log q$. A calibrated local-volatility field contributes the extra transport term $2\partial_k \kappa$, so the market skew enters the reverse dynamics twice: through the score of a skewed density and through the gradient of the noise geometry itself. More importantly, 6(i) makes the score available *without training*: since $q_T = \partial_k(e^{-k} \partial_k c)$ is produced by the same SPD solve as the price, $\partial_k \log q_T = \partial_k q_T / q_T$ is a post-processing of the state. E10 (10.9) measures the accuracy of that estimator against denoising and implicit score matching.

A P_1 finite-element function has no classical piecewise second derivative: its distributional second derivative is concentrated at nodes. E2 therefore recovers the financial density through the exact weighted-Dupire flux $q = e^{-k}(c_{kk} - c_k)$ on the output grid rather than interpreting $\partial_{kk} c_h$ as a classical finite-element field. Under input-matched noise, the correctly specified PDE fit has empirical mesh slope 0.64 with bootstrap interval $[0.60, 0.67]$, whereas applying the same differentiation directly to noisy quotes amplifies noise. We report this as a controlled empirical mesh study. A generic $O(h^2)$ theorem would require additional regularity and a dedicated mixed, flux-recovery, or dual Fokker–Planck discretisation. Non-negativity of the recovered density remains a diagnostic; SPD of the energy operator alone does not prove convexity preservation or complete static no-arbitrage.

8 Unified American pricing

American options replace the equation by an obstacle problem: with payoff Φ and discount/carry reaction r ,

$$\min(\partial_\tau u - \partial_k(\kappa \partial_k u) + r u, u - \Phi) = 0. \quad (23)$$

Backward Euler turns each maturity layer into the linear complementarity problem

$$\min_{u \geq \Phi} \frac{1}{2} u^\top \mathbf{A} u - b^\top u, \quad \mathbf{A} = \mathbf{M} + \Delta\tau(\mathbf{K} + r\mathbf{M}), \quad b = \mathbf{M} u^{\text{prev}}. \quad (24)$$

Since $\mathbf{A}$ is SPD (the reaction term adds a positive diagonal when $r > 0$), this is a *strictly convex* QP with a unique minimiser; we solve it by projected SOR and re-attach gradients through the fixed active set under strict complementarity, reusing the same stiffness matrix. The reaction term r is what creates a non-trivial early-exercise premium: with $r = 0$ (the driftless forward measure) an American call/put on a non-dividend forward is never exercised early, whereas $r > 0$ produces a genuine exercise region (5).

9 Statistical scope: prediction, recovery, and identifiability

A prediction-risk bound for the forward map and a coefficient-recovery bound for an inverse PDE are different statements. The latter requires not only a bounded smoothness class, for example

$$\mathcal{F}_s(R) = \{\kappa : \|\kappa\|_{H^s} \leq R, \kappa_{\min} \leq \kappa \leq \kappa_{\max}\},$$

but also an observation design, a noise model, and an inverse-stability or source condition linking price discrepancies to coefficient discrepancies. Discrete Fréchet differentiability alone does not supply this identification condition. Moreover, a field over (τ, k) is two-dimensional unless the estimator is explicitly decomposed into one-dimensional maturity slices.

For these reasons we do not state the previously suggested $N^{-s/(2s+1)}$ expression as a general theorem for Dupire coefficient recovery. E6 is instead an empirical scaling experiment under a

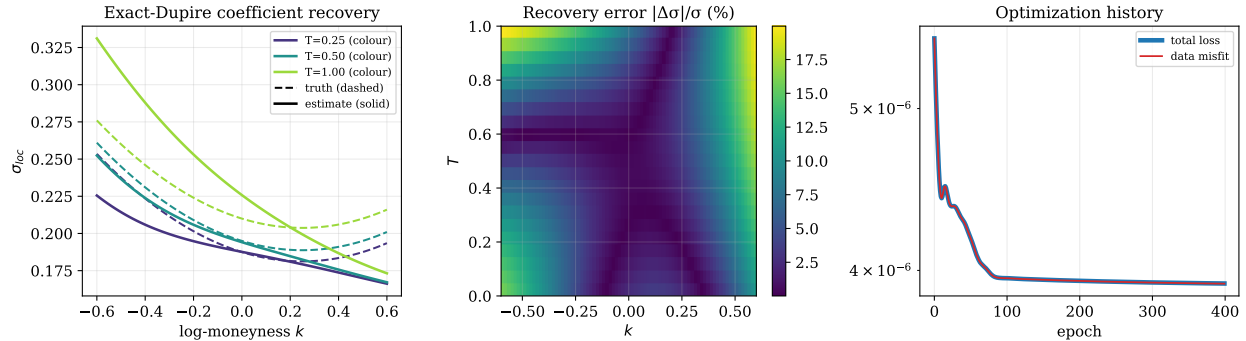


Figure 1: E1 exact-Dupire recovery on the working grid from an independently refined synthetic surface. Left: truth and estimate at selected maturities; centre: pointwise coefficient error; right: optimisation history for the final seed. Aggregate conclusions use all five stored replicates.

declared nested-sample protocol. Across 20 independent repetitions, its coefficient-RMSE slope is -0.42 (bootstrap interval $[-0.53, -0.31]$), while the error on an independent fixed price test set has slope -0.52 (interval $[-0.62, -0.40]$). These measured exponents are not compared to a direct-regression reference as evidence for a theorem. A valid inverse-problem theorem would additionally quantify the smoothing order of $\mathcal{Q}$, regularisation selection, and an inequality of the form

$$\|\kappa - \kappa^*\|_{\mathcal{X}} \leq C \|\mathcal{Q}(\kappa) - \mathcal{Q}(\kappa^*)\|_{\mathcal{Y}}^{\alpha}.$$

This distinction is substantive: the experiment describes observed sample-size behaviour of one implemented exact-Dupire design, not a dimension-free minimax law for a time-dependent local-volatility surface [19].

10 Experiments

Setup. All experiments use double-precision PyTorch on CPU, fixed seeds, and the Thomas solver. E1, E2 and E6 use the exact weighted-Dupire layer (5); E7 applies the same layer to normalized SPX calls. E3–E5 retain the symmetric benchmark (6) and are labelled as such. Synthetic coefficient studies use $\sigma(\tau, k) = 0.18 + 0.10k^2 - 0.05k + 0.03\tau$. E1 generates the clean surface on a 243×83 reference grid and inverts on an 81×21 grid, adds 1% noise, and uses an 80/20 observation split. It compares the neural field with a nodal Tikhonov inverse and an SVI-to-Dupire post-processing baseline. E2 feeds one nested realization of additive price noise to every method at each resolution and compares exact-Dupire flux differentiation with a smoothing spline, local polynomial, Tikhonov smoothing and raw finite differences; its headline values use additive normalized-price noise 10^{-4} . E3 uses paired seeds across noise levels; E4 uses a 60/20/20 train/validation/test split so that the reported μ is selected without access to the coefficient truth. E6 uses nested training prefixes and an independent fixed test set for every repetition. Every NPZ stores replicate-level arrays, while its JSON sidecar records configuration, seeds, source and data hashes, environment, run identifier and headline statistics.

The four additional studies use the same core implementation but different designs. E8 fixes the coefficient-network architecture, observation set, optimiser and step budget across four formulations. The hard branch maps its positive CoeffNet through the differentiable Galerkin solver, whereas the three soft branches additionally train a separate StateNet with a strong, Ritz or weak physics objective. Their state representation, regularisation and computational graphs therefore

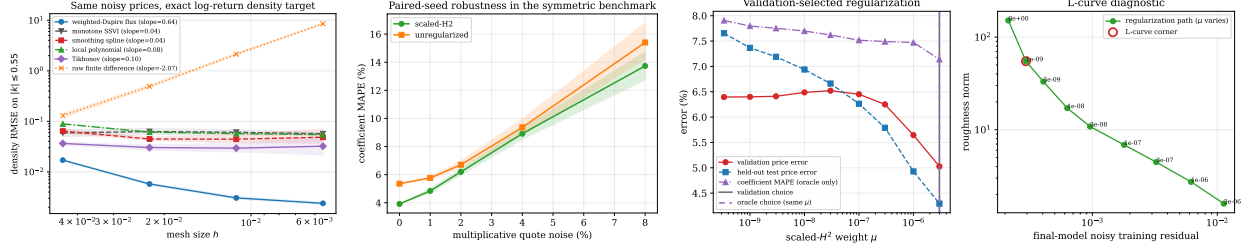


Figure 2: log-return- density RMSE from MAPE across identical noisy prices. Figure 3: coefficient noise levels. Figure 4: regularisation selection. Left: validation, held-out test and coefficient-oracle curves; right: the L-curve diagnostic.

differ, so E8 is a mechanism-level comparison rather than a fully controlled ranking. Each soft baseline’s physics weight is selected by an independent pilot using validation state-price MSE and then frozen. E9 draws smooth random coefficient fields, generates ground truth on a $4\times$ refined reference grid, and evaluates on four distribution-shift regimes. E10 uses a wider domain and a constant coefficient, so that the log-return density, its score and the exact reverse-time dynamics are all available in closed form, and reports Wasserstein distances relative to a Monte-Carlo self-noise floor. E11 places a spectral-decay Gaussian prior on a tensor Chebyshev coefficient basis and compares the Gauss–Newton–Laplace posterior against a parametric bootstrap, a Laplace-preconditioned random-walk sampler and a preconditioned Crank–Nicolson sampler. All computations are in double precision with fixed seeds, and every reported number is regenerated by the scripts listed in A.

10.1 E1 — Exact-Dupire synthetic coefficient recovery

The clean call surface is generated on a 243×83 grid and subsampled onto the distinct 81×21 inversion grid, avoiding an identical-grid inverse crime while retaining the same continuum model. Across five fixed noise realizations, the neural field attains coefficient MAPE 7.1% with a bootstrap interval $[5.1\%, 9.1\%]$ over 209 observations (152 train and 57 held out per realization). Its held-out price relative RMSE is 0.23%. The nodal Tikhonov inverse is stronger in this experiment: 5.2% coefficient MAPE and 0.20% held-out price error. Across the same five realizations, an SVI fit followed by numerical Dupire differentiation has 53.2% mean coefficient MAPE. Thus E1 verifies the exact-Dupire gradient and recovery pipeline but does not claim neural dominance over classical regularisation.

10.2 E2 — Same-noise financial-density recovery

At each of four nested grids ($N_k = 41, 81, 161, 321$), every estimator receives the same additive-noise realization of the same call prices. The target is the exact log-return density $q = e^{-k}(c_{kk} - c_k)$, not c_{kk} alone. The headline estimand fixes additive normalized-price noise at 10^{-4} and fits the preregistered $N_k = 81, 161, 321$ slope. Over 20 seeds, the correctly specified one-parameter Dupire fit reduces mean core RMSE from 1.71×10^{-2} to 2.39×10^{-3} , with empirical mesh slope 0.64 and bootstrap interval $[0.60, 0.67]$. On the finest grid, smoothing spline, local polynomial and Tikhonov estimates have RMSE 4.85×10^{-2} , 5.46×10^{-2} and 3.23×10^{-2} , respectively, while raw finite differences deteriorate to 8.55 and produce negative density on 45.5% of core nodes. This is now an input-matched comparison, but the PDE curve remains a correctly specified parametric fit whereas the smoothing methods are less restrictive. E2 therefore establishes stability in this controlled

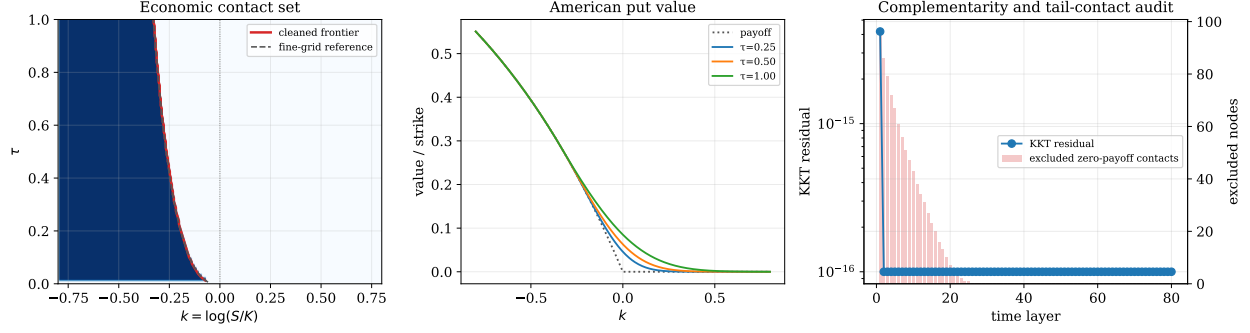


Figure 5: E5 American put audit. Left: cleaned economic contact set and refined reference frontier; centre: value and payoff slices; right: layerwise KKT residuals and excluded zero-payoff tail contacts.

model, not universal superiority over nonparametric density estimators or a P_1 second-derivative theorem.

10.3 E3–E4 — Noise robustness and deployable regularisation

E3 keeps the symmetric inverse-diffusion operator fixed and compares scaled- H^2 regularisation with the unregularised nodal field under paired noise realizations. At 8% multiplicative quote noise, ten paired seeds give coefficient MAPE 13.7% with empirical interval [12.7%, 14.7%] for the regularised estimator and 15.4% with interval [14.2%, 16.8%] without regularisation. The paired absolute improvement is 1.7%. These statistics quantify this declared benchmark; they do not imply that the gain must increase monotonically at every noise level.

E4 separates 60% training, 20% validation and 20% test observations for each of five seeds. Minimising mean validation price error selects $\mu = 3.0 \times 10^{-6}$ and yields held-out test price error 4.7%. The coefficient-truth oracle selects $\mu = 3.0 \times 10^{-6}$ with MAPE 7.1%; the validation choice has absolute oracle gap 0.00%. The independently computed L-curve corner is $\mu = 1.0 \times 10^{-9}$. Only the validation choice is deployable; the oracle and L-curve are reported as diagnostics.

10.4 E5 — Audited American contact frontier

With reaction $r = 0.05$, sparse Howard policy iteration is audited layer by layer against the primal, dual and complementarity conditions; PSOR is retained only as a post-solve diagnostic. Nodes where both value and payoff are numerically zero are excluded before constructing the economic contact set; up to 96 such tail contacts would otherwise be misclassified. The cleaned set is left-connected, contains 94 nodes at the longest maturity and contains 0 positive- k exercise nodes. Its frontier has RMSE 2.13×10^{-3} against an independently refined 641×161 Howard solve, while the maximum KKT residual is 4.19×10^{-15} . Thus the experiment validates the cleaned contact geometry and complementarity solve, not the discarded zero-payoff islands.

10.5 E6 — Nested-sample empirical scaling

For each of 20 independent repetitions, one randomized master sample is formed and the training sets are nested prefixes with $N \in \{20, 40, 80, 160, 320\}$. A disjoint fixed test set is held constant across all five prefix sizes within that repetition. The coefficient-RMSE log-log slope is -0.42 with bootstrap interval $[-0.53, -0.31]$; the corresponding independent-test price slope is -0.52 with

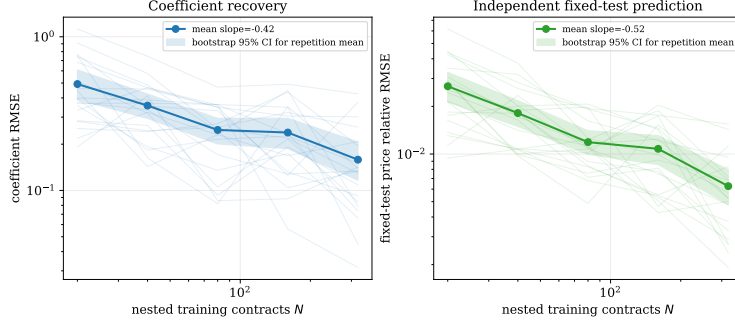


Figure 6: E6 nested-sample scaling over 20 repetitions. Left: coefficient RMSE; right: relative price RMSE on the independent fixed test set. Thin curves are individual repetitions and bands are replicate percentiles.

interval $[-0.62, -0.40]$. Both are empirical properties of this exact-Dupire design. We do not compare either exponent with a direct-regression rate or present it as an inverse-problem theorem.

10.6 E7 — Out-of-sample S&P 500 calibration

The delayed CBOE snapshot contains 16880 clean OTM quotes across 34 expiries, aggregated after grid snapping to 1534 unique price nodes. We compare the exact-Dupire field with monotone SSVI using held-out maturity blocks and contiguous held-out strike blocks, repeating every fold over the declared seeds. Pooling elementary OOF squared errors and counts before taking the square root gives maturity-holdout IV RMSE 2.29 volatility points for exact Dupire and 3.05 for SSVI; the strike-block values are 2.44 and 1.84, respectively. These split families test different extrapolation tasks and are reported separately rather than pooled across split families.

A final all-data fit is used only to display the coefficient and audit static shape diagnostics. Its liquid-core IV RMSE is 0.64 volatility points, the flux-density negative fraction is 0.0%, and the calendar-violation fraction is 0.0%. These diagnostics do not replace the held-out comparison and, more generally, SPD alone is not a complete static no-arbitrage certificate.

10.7 E8 — Hard constraints versus soft physics penalties

3 predicts that no finite penalty weight can certify admissibility. E8 probes that prediction against three network-based soft-constraint baselines. The methods share the coefficient-network family, observations, optimiser and training-step budget, but they do not differ only in penalty placement: the soft methods also learn a separate state network and use formulation-specific objectives. E8 is therefore a mechanism-level comparison, not a completely controlled architecture ranking.

Protocol. All four methods use the same CoeffNet MLP architecture for the diffusion coefficient on the same 65×33 evaluation grid, with the same Adam schedule (1200 steps, learning rate 2×10^{-3} and gradient clipping at 5), observation set and multiplicative quote noise. The methods differ as follows.

- **σ -Q (hard).** CoeffNet produces $\kappa_\theta = \text{softplus}(f_\theta) + \kappa_{\min}$ as in (9); the state is generated by the differentiable P_1 Galerkin/backward-Euler solver, with no independently learned StateNet or physics penalty. This branch also uses a coefficient-smoothness term.

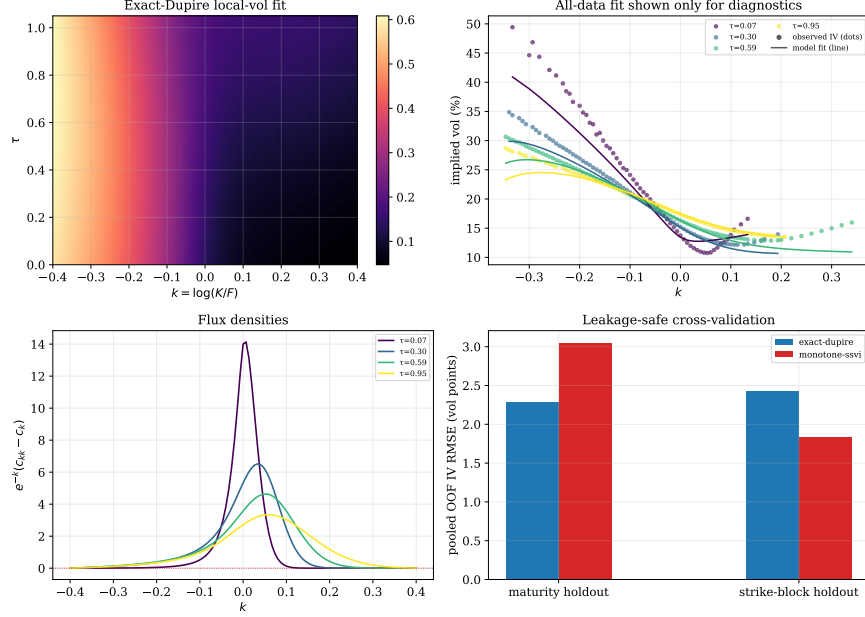


Figure 7: E7 S&P 500 study. Exact-Dupire local volatility and an all-data smile fit are shown only as diagnostics; the lower panels report flux densities and the cross-validated maturity/strike-block comparison with monotone SSVI.

- **PINN (soft).** CoeffNet and a separate StateNet MLP are trained jointly against observations, terminal/boundary penalties, positivity penalties and the strong-form collocation residual of (6).
- **Deep Ritz (soft).** The same two-network parameterisation is trained with an incremental backward-Euler variational energy in place of the strong residual.
- **VPINN (soft).** The same two-network parameterisation uses the residual tested against the P_1 hat basis, yielding a weak-form residual.

For each soft baseline, λ_{phys} is selected in an independent 400-step pilot at 2% noise by minimizing validation state-price MSE over the prespecified grid, and is then frozen for the reported runs. The tuning seed is excluded from the 5 evaluation seeds. Because the hard and soft branches differ in state representation, regularisation, parameter count, objective and computational graph, the comparison tests the operational consequences of hard versus finite-penalty enforcement rather than isolating a pure penalty-location effect.

Solver equivalence. Before comparing, we verify that our differentiable layer and a dense factorisation of the same assembled system agree to 2.8×10^{-16} in relative sup-norm, so any divergence between methods is attributable to the learning formulation and not to the linear algebra.

Result 1: admissibility. Across the recorded final fits, the hard architecture keeps $\min \kappa \geq 1.35 \times 10^{-2}$ for every evaluation seed and both noise levels, with zero negative interface coefficients by construction. All three soft baselines end with negative coefficients: $\min \kappa = -1.7 \times 10^{-3}$ (PINN), -2.0×10^{-3} (Deep Ritz), -9.0×10^{-4} (VPINN), with negative-interface fractions 6.9%, 55.4% and 8.8% of the grid. This pattern is consistent with the boundary-approach mechanism in

the proof of 3: a finite penalty can become small near the admissible boundary without certifying that the fitted coefficient remains inside it.

Result 2: two distinct failure modes. Consistent with 1, negative coefficients do not always destroy the layer. In the formal five-seed aggregate at 2% noise, the PINN systems have mean smallest-layer eigenvalue 2.0×10^{-2} , mean condition number 8 and mean Cholesky-failure count 0.0, whereas the σ -Q systems have $\lambda_{\min} = 2.50 \times 10^{-2}$ and condition number 7.2. A separate seed-11 profile supplies a deliberately local stress diagnostic: scanning all 32 implicit layers of that PINN fit finds 5 indefinite layers, with worst-layer eigenvalue -3.8×10^{-3} ; its middle layer still has $\lambda_{\min} = 9.1 \times 10^{-3} > 0$, so a single-layer audit would have missed the failure. In the same profile, the Deep Ritz and VPINN coefficients are negative but not negative enough relative to $-h/\Delta\tau_n$, so their layers remain SPD and the lumped mass masks the coefficient violation. This profile is evidence that the spectral failure can occur, not an estimate of its cross-seed frequency.

Result 3: the diagnostic that price fit hides. Every soft baseline fits its own price observations well: relative price RMSE 1.4% (PINN), 1.4% (Deep Ritz), 1.4% (VPINN) versus 0.58% for σ -Q. Yet coefficient recovery separates sharply: 4.3% MAPE at 2% noise and 6.7% at 8%, versus 40.3%, 80.9% and 50.5%. The learned PINN state also violates the sampled-grid strike-monotonicity diagnostic on 12.3% of tested node pairs. This diagnostic is not a complete static-arbitrage certificate, but it shows why low residual on the observed functional says little about either the coefficient or untested shape restrictions.

Result 4: transfer as an operational test. We take each learned coefficient and feed it to the *same* SPD Galerkin propagator, then measure the relative price error on a held-out maturity block. For the hard architecture the transfer error coincides with its own fit error, 0.58%, because its state *is* the propagated Galerkin solution; the transfer succeeds on 100% of seeds. All formal PINN transfers are factorable, but their mean held-out error reaches 6.0×10^{-2} at 8% noise; VPINN reaches 6.0×10^{-2} , while Deep Ritz degrades to 28.1% already at 2%. Gradient magnitudes tell the same story: the largest coefficient-gradient norm during training is 2.0×10^{-4} for σ -Q against 0.13 for PINN and 0.1 for VPINN.

10.8 E9 — Zero-shot generalisation against neural operators

E9 evaluates the amortisation question of C.2: viewed as a map from a coefficient field to a price surface, how does the σ -Q operator compare with learned operators under distribution shift in the strike grid and the maturity horizon?

Protocol. We sample coefficient fields from a smooth random family using one fixed data seed, generate ground-truth surfaces on a $4 \times$ refined reference grid, and restrict them to the working grid. Learned operators see 640 training instances and are evaluated on 160 held-out instances in four regimes: **R1** in-distribution (65×33), **R2** strike refinement (129×33), **R3** maturity refinement (65×49), and **R4** horizon extrapolation ($\tau_{\max}$ extended from 0.8 to 1.2). A 2-D FNO, a DeepONet and a graph kernel/neural operator are each trained from five paired initialisation seeds on the same data split, with a fixed 120-epoch cap and the best validation checkpoint retained. Their parameter counts, architecture-specific batch sizes and realised training times differ, so E9 reports cross-seed performance under a common dataset and stopping rule rather than claiming equal compute or convergence. The σ -Q operator has *no trainable parameters* in this experiment: it simply assembles and solves.

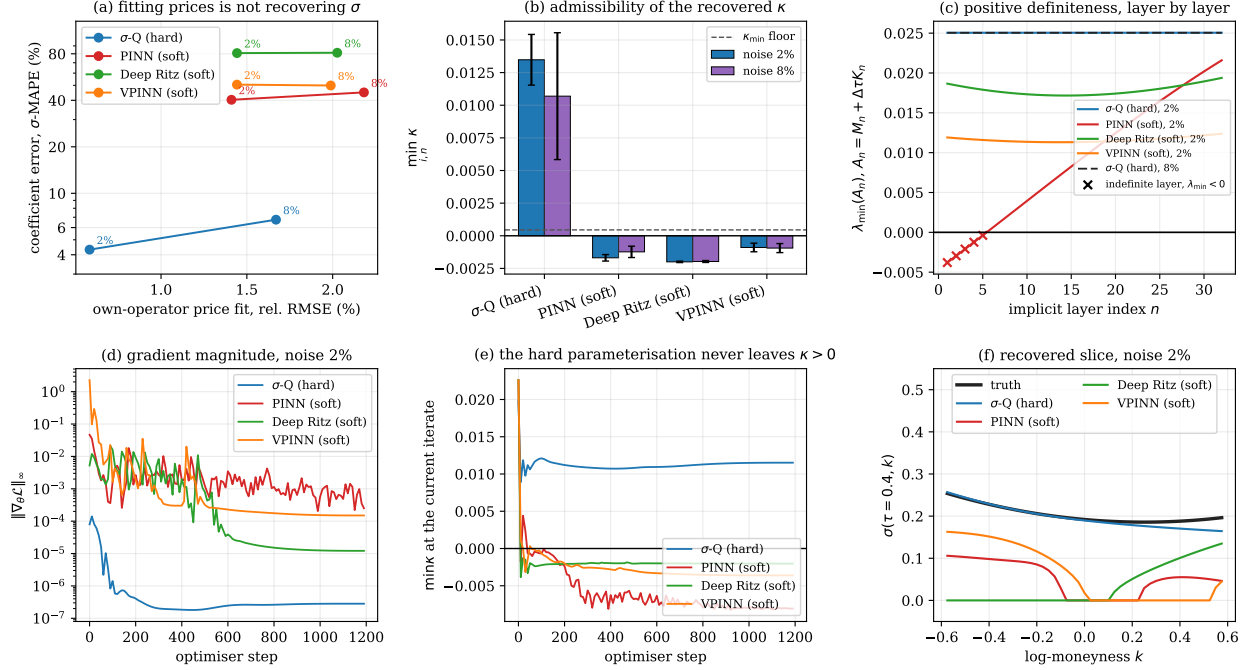


Figure 8: E8 hard versus soft constraints. Coefficient MAPE and minimum interface coefficient (top left, top centre) summarize 5 formal seeds. The per-layer spectrum and optimization traces are the separate seed-11 profile: its PINN fit has 5 indefinite layers that a single-middle-layer audit would miss. The recovered slice completes the mechanism diagnostic; the profile panels are not cross-seed frequency estimates.

Result 1: an honest in-distribution loss. In R1 the FNO beats us: relative L^2 error 1.93×10^{-3} versus 4.36×10^{-3} for σ -Q, with DeepONet at 5.99×10^{-3} and the graph operator at 7.74×10^{-3} . The reason is instructive rather than embarrassing. The learned operators are trained against the refined reference, so they learn to *correct* the coarse-grid discretisation error that σ -Q, being a consistent scheme, faithfully commits. Reporting this inversion is essential: hard constraints are not free accuracy.

Result 2: the crossover under refinement. Under strike refinement (R2) the ordering inverts. Our error *falls* to 2.33×10^{-3} , because a finer mesh is simply a better discretisation, while the FNO *rises* to 3.53×10^{-3} , because its learned correction was calibrated to the training resolution. This is the operational meaning of discretisation convergence versus discretisation invariance: the latter is a representation property, not a guarantee of accuracy transfer. Under maturity refinement (R3) the two are comparable (4.33×10^{-3} versus 4.09×10^{-3}).

Result 3: horizon extrapolation. R4 is the decisive regime. σ -Q degrades gracefully to 6.12×10^{-3} while the FNO degrades to 3.59×10^{-2} , approximately $5.9 \times$ the σ -Q error, with DeepONet at 1.19×10^{-2} and the graph operator at 1.61×10^{-2} . The failure is structural, not merely numerical: the FNO output is negative on 21.6% of nodes (already 13.7% in R1) and violates monotonicity in k on 19.0%. The corresponding sampled-grid negativity and monotonicity diagnostics are zero for the σ -Q surface in all four regimes. These measured zero fractions are empirical diagnostics, not a certificate of complete static no-arbitrage.

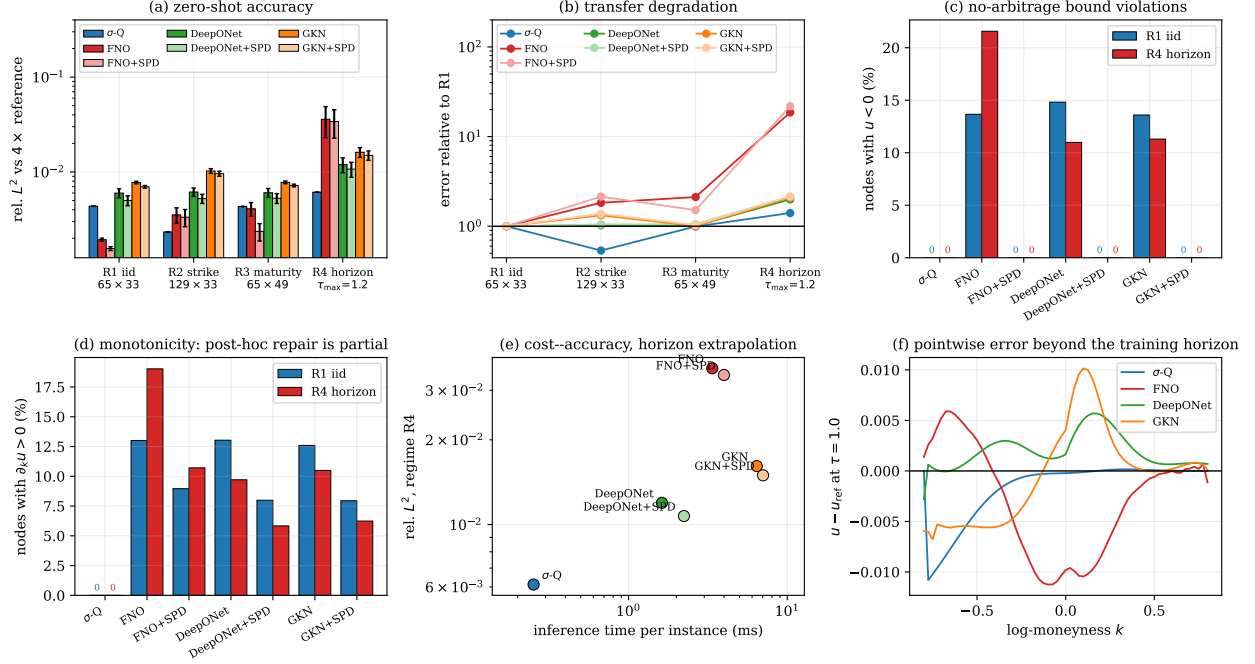


Figure 9: E9 zero-shot generalisation, 160 held-out instances per regime and five paired training seeds for each learned operator. Points summarize cross-seed means and intervals summarize the corresponding 95% uncertainty. The FNO wins in-distribution (R1) but loses under strike refinement (R2) and degrades sharply on horizon extrapolation (R4), where its sampled outputs are also negative and non-monotone; the σ -Q operator records zero violations of these sampled-grid diagnostics and the lowest per-instance inference cost.

Result 4: post-hoc repair is not a hard constraint. Applying one post-hoc σ -Q/SPD correction step to each learned surface (the +spd variants) reduces the targeted box violations, but it is not a projection onto the complete admissible set and therefore supplies no static- arbitrage certificate. It still leaves 10.7% sampled-grid monotonicity violations for FNO on R4 and does not restore accuracy. A single repair pass addresses a measured symptom, not the learned operator.

Result 5: cost. Per-instance inference takes 0.16 ms for the tridiagonal sweep against 2.4 ms for the FNO and 4.6 ms for the graph operator, and the learned operators additionally require 4.5×10^5 parameters and 460 s (FNO), 219 s (DeepONet) or 997 s (graph operator) of training. Amortisation is therefore not automatically the cheaper route for a banded one-dimensional operator; it becomes attractive when the inner solve is expensive, which is precisely the hybrid regime described in C.2.

10.9 E10 — The dual density, its Fokker–Planck flow, and score models

E10 tests the three claims of 6 numerically and then uses the resulting operator as a *training-free* score estimator against score-matching and DDPM baselines.

Protocol. We work on a wider domain ($|k| \leq 1.6$) with a 129×65 working grid and a 513×257 reference grid, and use a constant benchmark coefficient so that the exact log-return density, its score and the exact reverse-time dynamics are available in closed form. All errors are measured against those closed forms; Wasserstein distances are reported relative to a Monte-Carlo self-noise

floor computed from independent draws of the exact law, so that “at the floor” is a meaningful statement.

Result 1: convergence and structure. Under joint refinement the state converges with empirical spatial slope 1.73 and the dual density with slope 1.49; in time both converge at 0.98 and 1.00, the expected first order for backward Euler. Every one of the implicit layers is SPD ($\lambda_{\min} = 0.191$, largest condition number 64.9, zero factorisation failures).

Result 2: mass conservation is a discretisation invariant, not a time-stepping artefact. 6(ii) predicts $\int q_T = 1$ exactly in the continuum. The measured defect $|\int q - 1|$ falls from 4.2×10^{-4} to 1.6×10^{-6} when the mesh in k is refined with the time grid held fine, but stays essentially flat at 4.1×10^{-7} when only $\Delta\tau$ is refined with the mesh in k held fine. The defect is therefore controlled by the spatial boundary bracket in the proof, exactly as the analysis says, and refining the time step alone cannot remove it. The Fokker–Planck residual of (19) converges at empirical slope 1.12.

Result 3: the forward SDE. Sampling (20) by Euler–Maruyama and comparing with the operator density gives $W_1 = 4.7 \times 10^{-4}$, 6.2×10^{-4} and 5.1×10^{-4} at 60, 240 and 960 steps, against a Monte-Carlo self-distance floor of $5.7 \times 10^{-4} \pm 1.5 \times 10^{-4}$ for the same particle count. All three lie within one standard deviation of the floor, and the constant- κ control against the exact Gaussian gives 4.8×10^{-4} on the same footing. The comparison is therefore sampling-noise limited rather than discretisation limited, which is why the three values are not monotone in the step count; what it establishes is the sign and form of the drift, namely that $-\kappa$ in (20), and not a zero-drift schedule, is the one consistent with the martingale normalisation. Resolving a difference below 1.5×10^{-4} would require more particles, not finer steps.

Result 4: the operator is a zero-training score model. The score $\partial_k \log q$ obtained by differentiating the operator density attains relative L^2 error 2.06×10^{-2} with *no training at all*. Denoising score matching, trained to convergence in 30 s per fit, reaches 9.5×10^{-2} at the largest sample size and 0.45 at the smallest; implicit (Hyvärinen) score matching, at 78 s per fit, reaches only 0.92. The gap is the expected bias–variance contrast between the two estimators, and both are dominated by simply differentiating a structure-preserving forward solve when such a solve exists.

Result 5: reverse-time sampling. Plugging each score into the Anderson reverse-time drift of 7 and sampling, the operator score reaches W_1 equal to $0.74\times$ the Monte-Carlo floor on the fine grid and $3.3\times$ on the working grid, versus $6.2\times$ for the best DSM model, $16.4\times$ for the best ISM model (up to $2.3 \times 10^3\times$ at small sample size) and $1.67\times$ for a tuned variance-preserving DDPM. Wall-clock is 6.9 ms for the operator route against 42 s of training plus sampling for DDPM. We state the scope precisely: this is a one-dimensional analytically tractable density, so the result does not claim superiority of Galerkin scores in high-dimensional generative modelling. It does establish that the market-calibrated diffusion field of (20) is a genuine, mass-conservative score model rather than a loose analogy.

10.10 E11 — Laplace uncertainty quantification against bootstrap, MCMC and importance sampling

Protocol. This study executes the programme of C.6 on synthetic truth, where the coefficient field is known pointwise and coverage is therefore measurable.

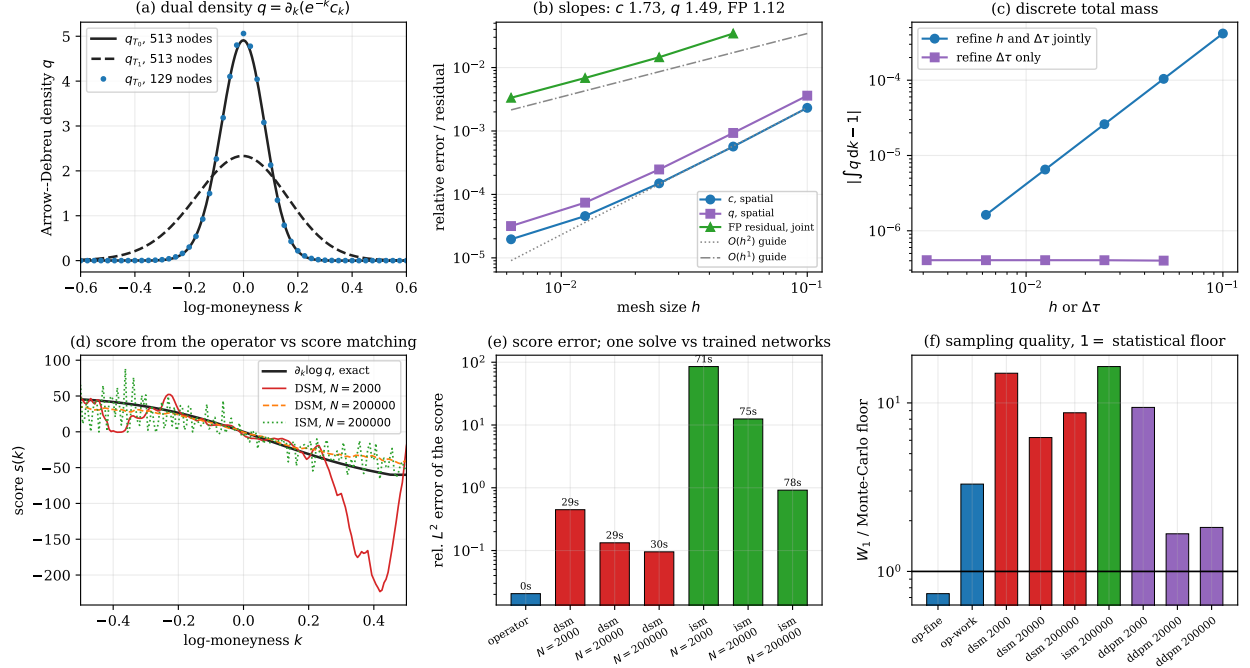


Figure 10: E10 dual density and score models. Top: mesh convergence of state and dual density, the mass defect under joint versus time-only refinement, and the Fokker–Planck residual. Bottom: forward-SDE agreement at the Monte-Carlo floor, score error for the operator versus DSM/ISM at three sample sizes, and reverse-time sampling error relative to the floor including a DDPM baseline.

- *Parameterisation and prior.* The pre-softplus field f is a tensor Chebyshev expansion of degree 3 in τ and 8 in k , so $\dim \Theta = 36$, and $\kappa = \kappa_{\min} + \text{softplus}(f)$ keeps every candidate admissible by construction. The prior is the proper Gaussian $\mathcal{N}(m_0, C_0)$ with spectral-decay scales $s_{mn} = s_0 / ((1+m)^A(1+n)^A)$, $s_0 = 2$, $A = 1.5$, and m_0 the constant field $\sigma \equiv 20\%$. This is a genuine $C_0 \succ 0$, not an H^2 seminorm, so 8 applies verbatim.
- *Data.* 240 noisy call quotes on a τ - k grid with $|k| \leq 0.6$, generated from a 257×129 reference solve and multiplicative noise of 2% or 8%; the working solve is 65×33 , so the study carries a deliberate discretisation mismatch. Three truths are used (**paper**, **skewed**, **bumpy**).
- *Point estimate and Laplace covariance.* Adam (900 steps) then L-BFGS (250 iterations) then Levenberg–Marquardt-damped Gauss–Newton, to a gradient norm of 5.4×10^{-10} ; J is the exact 240×36 tangent Jacobian assembled from 5 in 0.03s, and $\Sigma = H_{\text{GN}}^{-1}$ from (28). Bands for σ_{loc} and for the dual density rows at $\tau \in \{0.2, 0.4, 0.8\}$ use the delta method (29).
- *Four reference computations.* (i) A parametric bootstrap with $B = 200$ noise resamples, each refitted to convergence with a per-replicate Gauss–Newton polish (worst gradient 2.1×10^{-5}), which targets the sandwich covariance S of (30); (ii) the analytic sandwich S itself; (iii) two Metropolis samplers on the same posterior — a Laplace-preconditioned random-walk Metropolis (pRWM) and a preconditioned Crank–Nicolson (pCN) chain, each 4 chains of 9000 draws after 3000 burn-in, with split- $\hat{R}$ and initial-positive-sequence ESS diagnostics; (iv) self-normalised importance sampling with the Laplace proposal, reporting weight ESS.
- *Frequentist coverage.* Independently of the single-realisation study, six configurations (three

truths $\times$ two noise levels) are each refitted under $R = 24$ fresh noise draws, and pointwise 95% coverage of the true surface is evaluated on a 1,363-node grid, giving 32,712 coverage tests per configuration, summarised with Wilson intervals.

Results.

1. *The bootstrap matches the sandwich, not the posterior, exactly as predicted.* At 2% noise the bootstrap-to-Laplace standard-deviation ratio has median 0.61 (p10 0.35, p90 0.87), i.e. the bootstrap is systematically *narrower*. Against the analytic sandwich the same replicates give median 1.00 (p10 0.96, p90 1.05), and the purely analytic shrinkage — the median over evaluation nodes of the sandwich-propagated σ standard deviation divided by the Laplace-propagated one, computed without any resampling — is 0.606. The three numbers are mutually consistent to two digits and reproduce across all four configurations (bootstrap/Laplace medians 0.61, 0.61, 0.68, 0.65; bootstrap/sandwich medians 1.00, 1.00, 1.00, 0.95). The discrepancy is a scale factor and not a shape disagreement: the pointwise correlation between the bootstrap and Laplace standard-deviation fields is 0.947 at 2% noise, and the same shrinkage survives the delta-method push-forward to the Arrow–Debreu density, where the bootstrap/Laplace ratio has median 0.71. Without 8 the first ratio would read as a 39% failure of the Laplace approximation; with it, the study instead confirms both objects.
2. *Two independent posterior samplers reproduce the Laplace spread.* The pRWM chains mix acceptably (acceptance 0.22, split- $\hat{R}$ max 1.030 and median 1.013, ESS min 225 and median 325) and give a sampler-to-Laplace standard-deviation ratio of median 1.04 (p10 0.99, p90 1.06) with pointwise correlation 0.9995 across the whole surface; the posterior mean sits 0.070 posterior standard deviations from the MAP in median and 0.224 at worst, so the Gaussian centring is also accurate. Self-normalised importance sampling with the Laplace proposal is even sharper: weight ESS 96% of draws at 2% noise (81% at 8%), ratio median 1.003, correlation 0.9996. A high weight ESS is direct evidence that the Laplace density is close to the posterior in the relevant f -divergence, not merely in its second moment.
3. *Coverage: honest, informative, and mildly conservative under noise.* On a single realisation the 95% Laplace band contains the true σ_{loc} at 89.1% of nodes and the true density at 88.7%; the pRWM band gives 89.1% on the same realisation, so the shortfall is a property of the posterior and of the discretisation mismatch, not of the Gaussian approximation. Under repeated noise the frequentist coverage is 93.4% (Wilson [93.2%, 93.7%]) for the reference configuration and ranges over [87.9%, 97.6%] across the six configurations, with mean band width growing from 0.080 to 0.107 as noise grows from 2% to 8%. The under-coverage cases are the rough **bumpy** truth at low noise, where a degree-(3, 8) Chebyshev field cannot represent the truth and the residual bias is not a variance; the over-coverage cases are the high-noise runs, where the bands are wide enough to absorb that bias. This is the expected bias-variance signature of a regularised inverse problem and we report it as such rather than claiming nominal calibration.
4. *Uncertainty amplifies in the wings, and every posterior draw stays admissible.* The pointwise standard deviation of σ_{loc} spans $[8.3 \times 10^{-4}, 6.9 \times 10^{-2}]$ over the evaluation grid, a factor of 83, rising from ≈ 0.001 at the money and the shortest maturity to ≈ 0.045 – 0.049 at $|k| = 0.575$; this is the correct qualitative response to a design that places no quotes beyond $|k| = 0.6$. Structural audits of the draws confirm that uncertainty never buys inadmissibility: over all 200 bootstrap refits of the reference configuration the minimum interface coefficient is 8.3×10^{-3}

with zero Cholesky failures, layer $\lambda_{\min} \geq 0.45$, recovered density mass in $[0.99985, 1.00003]$ and $q_{\min} \geq -1.1 \times 10^{-12}$; over the sampled posterior draws of all four configurations the same audit gives zero Cholesky failures, q negativity on at most 0.28% of nodes, and $q_{\min} \geq -1.6 \times 10^{-12}$, i.e. at the level of double-precision round-off rather than a genuine sign violation. At the MAP itself the 36-dimensional Gauss–Newton Hessian is well conditioned enough to invert densely ($\text{cond } H_{\text{GN}} = 3.4 \times 10^5$), all 32 implicit layers are SPD with $\lambda_{\min} = 0.714$ and $\text{cond} \leq 14.4$, the recovered mass is 1.000015, and the fit reaches σ -MAPE 7.6% at price rel-RMSE 0.46%. Raising the quote noise to 8% moves all four diagnostics in the direction the theory predicts and none of them in the wrong one: the point estimate degrades to σ -MAPE 12.7%, the Gauss–Newton Hessian becomes *better* conditioned ($\text{cond } H_{\text{GN}} = 2.4 \times 10^4$, because the prior curvature now dominates a weaker likelihood), the mean 95% half-width widens from 0.0397 to 0.0511, and the single-realisation coverage rises from 89.1% to 91.6%. The bands are therefore not calibrated at one noise level and extrapolated; they respond to the information content of the data.

5. *A negative control we retain deliberately.* The pCN chain, although it accepts at a healthy rate, fails to mix on this posterior: split- $\hat{R}$ reaches 33.5 with ESS ≈ 47 , and it reports a spread ratio of 1.45 against Laplace. Comparing the two samplers directly, the σ -standard-deviation fields disagree by a median of 41% and up to 202%. We keep this in the record because it shows what the diagnostics are for: had we run pCN alone and skipped $\hat{R}$, we would have reported a 45% “correction” to the Laplace bands that is in fact a mixing artefact. The prior-centred pCN proposal is poorly matched to a likelihood-dominated posterior whose Hessian has condition number 3.4×10^5 ; pRWM, which preconditions with Σ , does not have this problem.
6. *Cost.* The complete Laplace pipeline — MAP fit 6.2s plus Jacobian 0.03s plus a dense 36×36 factorisation — is three orders of magnitude cheaper than its own validation (204s for the bootstrap, 239s for pRWM per configuration; 3943s for the whole study). This is the practical argument for the adjoint-based Laplace route: the validation is expensive, the deliverable is not.

11 Discussion and limitations

The strongest result is an architectural one: positivity and SPD are enforced before optimisation, while smoothness and data fit remain tunable statistical choices. 3 shows that this is a difference in kind rather than degree, and E8 observes the predicted failure patterns in a mechanism-level comparison whose state representations and objectives are not fully controlled. Several limits matter, and we state them without softening.

First, E1, E2, E6 and E7 use the exact weighted-Dupire operator (5), whereas E3–E5 and E8–E11 deliberately retain the symmetric coefficient-diffusion benchmark (6) to isolate regularisation, obstacle, constraint, operator-learning, score and UQ questions. Numbers from the latter group are therefore not exact local-volatility evidence. Second, SPD neither implies a uniformly small condition number nor guarantees convexity preservation, density non-negativity, or complete static no-arbitrage; 1 makes the first gap precise, while E7 reports rather than assumes the flux-density and calendar diagnostics (0.0% and 0.0% violations on the sampled grids). A zero sample fraction, when obtained, is still an empirical diagnostic rather than a certificate. Third, the E2 density is a post-processing diagnostic and the E6 sample-size slopes are empirical rather than a minimax theorem. Fourth, gradients through an American QP are locally valid under a fixed active set

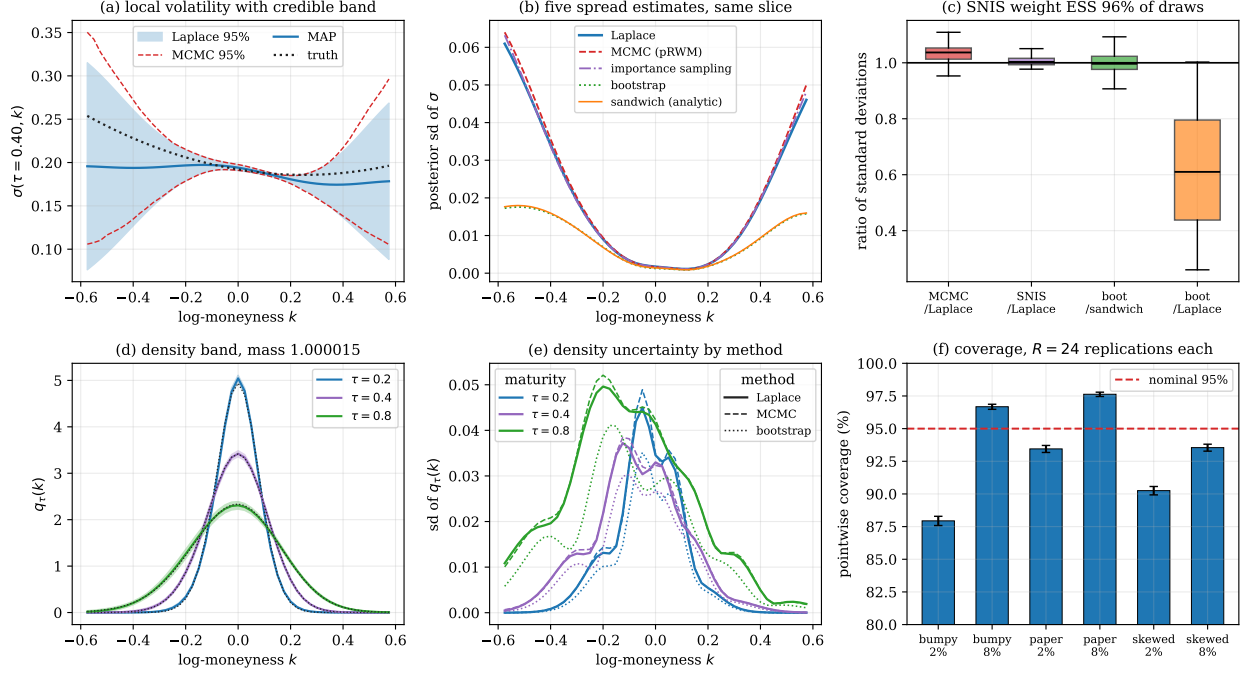


Figure 11: E11 uncertainty quantification. Top: a local-volatility slice with Laplace and MCMC credible bands; the five spread estimates on that slice, where Laplace, pRWM and importance sampling coincide while the bootstrap tracks the analytic sandwich; and the distribution of the four ratios over the whole surface. Bottom: Arrow–Debreu density bands with recovered mass, the density spread by method, and repeated-noise coverage with Wilson intervals for six truth/noise configurations against the nominal 95% line.

and strict complementarity; exercise-boundary changes create nonsmooth points, and E5 therefore reports both KKT residuals and the rule used to remove numerical zero-payoff tail contacts.

Three further limits belong specifically to the new studies. E9 shows that hard constraints cost in-distribution accuracy: a trained FNO is roughly $2.4\times$ more accurate than the solver on the training grid, and only becomes worse under refinement or extrapolation. Reporting the crossover rather than the favourable regime alone is the honest presentation, and it also identifies the regime in which amortisation genuinely pays—an expensive inner solve, which this banded one-dimensional operator is not. E10 is a one-dimensional benchmark with a closed-form score, so it establishes the exactness of the dual view and the viability of a training-free score, not any claim about high-dimensional generative modelling. E11 validates the Laplace approximation against independent references, but the validation is local by construction: the Gauss–Newton Hessian is evaluated at a single mode, and the sandwich identity of 8 shows that the posterior spread and the frequentist spread of the penalised estimator answer different questions, so the two must not be conflated even when both are correctly computed.

These limitations define a revised priority order. The immediate empirical step is external validation of the exact-Dupire studies: multiple independent market dates for E7, wider quote regimes, and prespecified tuning rules rather than reuse of one snapshot. Migrating E3–E5 to weighted Dupire would answer a different financial question and should be done only with a re-designed loss, obstacle and diagnostic protocol, not by relabelling the symmetric numbers. Beyond that, the entropic martingale bridge and the rough or nonlocal operators change the model class

or state space, and each warrants its own feasibility and identification study rather than an ablation. A higher-dimensional score experiment, in which the inner solve is expensive enough for amortisation to pay, is the natural continuation of E9 and E10 together.

12 Conclusion

We presented a hard-constrained differentiable Galerkin architecture for inverse diffusion learning and used local-volatility calibration as its motivating test bed. The neural component parameterises a positive coefficient; the numerical component enforces a coercive evolution through SPD implicit layers. This split supports adjoint calibration and a convex obstacle extension while making clear which properties are architectural, which are statistical, and which remain empirical diagnostics. We proved that no finite soft penalty can certify admissibility, that discrete layer SPD is strictly weaker than coefficient positivity, that the coefficient-to-state map is Fréchet differentiable with an explicit adjoint, and that the Dupire dual is exactly the Fokker–Planck equation of a log-forward diffusion with an Anderson reversal. We then tested those claims against soft-constrained PINN, Deep Ritz and VPINN formulations, against FNO, DeepONet and graph neural operators under distribution shift, against denoising and implicit score matching and a DDPM, and against a parametric bootstrap and two MCMC samplers. The evidence supports the architectural thesis outside the training distribution and, equally clearly, shows where a learned operator is more accurate inside it. The exact weighted Dupire formulation and the symmetric benchmark remain explicitly separated, and entropic martingale transport together with rough or nonlocal dynamics remain mathematically specified interfaces rather than completed experiments. The broader lesson is that in inverse PDE learning, neural flexibility can be confined to the unknown coefficient while numerical structure remains in control.

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A Technical scope and reproducibility notes

Status of each formal statement. Complete, self-contained finite-dimensional proofs are given for 2 (SPD from positive interface weights), 3 (no-go for soft penalties), 4 and 1 (layer spectrum and the strict gap between layer SPD and admissibility), 5 (discrete Fréchet differentiability and the adjoint recursion, by the implicit function theorem), 6 (dual density, unit mass, and the forward equation) and 8 (sandwich covariance). 1 invokes standard non-autonomous coercive-form parabolic theory; 7 invokes Anderson’s reverse-time theorem [20] and supplies only the algebra of the state-dependent case. We make no theorem-level claim for the recovered-density mesh slope, the E6 sample-size slopes, or the E9/E10 empirical convergence rates: these are measured quantities. The exact local-volatility interpretation applies to E1, E2, E6 and E7, which execute (5); E3–E5 and E8–E11 retain the symmetric operator and are interpreted only under their explicit benchmark protocols.

Software and determinism. All experiments use double-precision PyTorch [44] on CPU, with fixed seeds recorded in each script. Optimisation uses Adam [42] throughout, L-BFGS [43] for the E11 MAP polish, and Levenberg–Marquardt-damped Gauss–Newton [41] for each bootstrap replicate. Linear layers are solved by the Thomas algorithm; every reported spectral quantity is computed by dense symmetric eigendecomposition of the assembled layer matrices, not estimated.

Experiment-to-artefact map.

Study	Script	Result file	Figure
E1–E2	src/experiments.py	results/E1_recovery.*; results/E2_density.*	1–2
E3–E4	src/experiments.py	results/E3_noise.*; results/E4_lambda.*	3–4
E5–E6	src/experiments.py	results/E5_american.*; results/E6_generalization.*	5–6
E7 SPX	src/run_spx.py	results/SPX_calib.*	7
E8 hard vs. soft	src/ext/E8_structure.py	E8_structure.json	8
E8 layer profiles	src/ext/E8b_profiles.py	E8b_profiles.json	8
E9 operators	src/ext/E9_operator.py	E9_operator.json	9
E10 diffusion	src/ext/E10_diffusion.py	E10_diffusion.json	10
E11 UQ	src/ext/E11_uq.py	E11_uq.json	11
shared core	src/ext/common.py	—	—
figures	src/make_figures.py; src/ext/make_figures_ext.py	—	—

Approximate single-run wall-clock on a laptop CPU is 515s for E8, 1998s for E9, 799s for E10 and 3943s for the full E11 sweep, the latter comprising four configurations of MAP-plus-Laplace (≈ 6.2 s each), bootstrap (≈ 204 s each), pRWM (≈ 239 s each) and pCN (≈ 400 s each), plus six coverage configurations at ≈ 90 s each.

E1–E7 commands and full configurations. E1–E6 are run individually as `python3 -u src/experiments.py EX --mode full`, replacing EX by E1,...,E6; E7 uses `python3 -u src/run_spx.py --mode full`. E1 uses five seeds and 400 steps on an 81×21 inverse grid with a 243×83 generator. E2 uses 20 seeds and four nested strike grids. E3 uses ten paired seeds, five noise levels and 300 steps. E4 uses five seeds, nine candidate smoothness weights, 300 steps and a 60/20/20 split. E5 uses three working grids, 81×21 , 161×41 and 321×81 , against a 641×161 Howard reference. E6 uses 20 primary repetitions plus 45 declared sensitivity runs, five nested sample sizes and 250 primary steps. E7 uses a 121×33 grid, five folds in each split family, three seeds and 500 steps. After all sidecars exist, the deterministic order is `src/ext/update_e1e7_macros.py`, then `src/make_figures.py`, followed by

Experiment	Metric	σ -Q / exact Dupire	Baseline or diagnostic
E1 coefficient recovery	mean MAPE	7.1%	nodal Tikhonov 5.2%
E1 held-out prices	rel. RMSE	0.23%	nodal Tikhonov 0.20%
E2 density, finest grid	RMSE	2.39×10^{-3}	Tikhonov 3.23×10^{-2}
E3 noise (8%)	coefficient MAPE	13.7% (reg.)	15.4% (no reg.)
E4 deployable selection	validation μ / test error	3.0×10^{-6} / 4.7%	oracle $\mu = 3.0 \times 10^{-6}$
E5 obstacle solve	frontier RMSE / KKT	2.13×10^{-3} / 4.19×10^{-15}	fine-grid Howard reference
E6 nested sample scaling	coefficient slope	-0.42	CI $[-0.53, -0.31]$
E7 maturity holdout	pooled OOF IV RMSE	2.29	SSVI 3.05
E7 strike-block holdout	pooled OOF IV RMSE	2.44	SSVI 1.84
E8 hard vs. soft	coefficient MAPE	4.3%	PINN 40.3%
E8 admissibility	min κ / neg. fraction	1.35×10^{-2} / 0%	PINN -1.7×10^{-3} / 6.9%
E8 seed-11 profile	indefinite layers	0/32	PINN 5/32
E9 zero-shot (in-dist.)	rel. L^2	4.36×10^{-3}	FNO 1.93×10^{-3}
E9 zero-shot (horizon)	rel. L^2	6.12×10^{-3}	FNO 3.59×10^{-2}
E9 structural violations	neg. + non-monotone	0% (all regimes)	FNO 21.6% / 19.0%
E10 dual density	mass defect $ \int q - 1 $	1.6×10^{-6}	—
E10 score error	rel. L^2 , no training	2.06×10^{-2}	DSM 9.5×10^{-2}
E10 reverse sampling	W_1 /MC floor	0.74×	DDPM 1.67×
E11 Laplace UQ	sd ratio vs. reference	1.04 (MCMC)	0.61 (bootstrap)
E11 coverage	95% band coverage	89.1%	MCMC 89.1%

Table 1: Summary of experimental results. E1, E2 and E6 use the exact weighted Dupire layer, E3–E5 are explicitly symmetric inverse-diffusion benchmarks, and E7 is a single-snapshot SPX study with maturity and contiguous-strike holdouts. The outcomes are mixed rather than uniformly favourable: nodal Tikhonov has lower coefficient error in E1, and E7 reports exact-Dupire and monotone-SSVI holdout errors side by side rather than selecting by in-sample fit. E8–E11 test separate structure, amortisation, diffusion and uncertainty claims.

`src/ext/audit_numbers.py` and `src/ext/audit_macros.py`. The macro updater first rejects any non-full sidecar and atomically rebuilds `summary_E1_E7.json`; the audits check rounding, labels, full-mode provenance, seed counts and NPZ/JSON metadata identity.

Baseline configurations. E8 uses 1200 Adam steps at learning rate 2×10^{-3} with gradient clipping at 5, hard-branch smoothness weight $\mu = 10^{-5}$, seeds $\{11, 23, 37, 53, 71\}$ and noise levels $\{2\%, 8\%\}$. An excluded pilot seed selects $\lambda_{\text{phys}} = 10^{-2}$ for PINN, Deep Ritz and VPINN from the grid $\{10^{-2}, 10^{-1}, 1, 10, 100, 1000\}$ by validation state-price MSE. E9 trains for 120 epochs under architecture-specific batch sizes: the FNO has width 24, depth 4 and (8, 12) retained modes (4.5×10^5 parameters), the DeepONet a latent dimension of 128, and the graph operator width 24 with spatial and temporal radii 0.075 and 0.20. E10 uses a 129×65 working grid against a 513×257 reference, a constant $\sigma = 0.25$, 2×10^5 Monte-Carlo paths, 480 reverse steps and score clipping at 60. E11 uses a 3×8 tensor Chebyshev basis (36 parameters), prior scale $s_0 = 2$ with decay exponent $A = 1.5$, 240 synthetic contracts, 200 bootstrap replicates, and four chains of 9000 draws with 3000 burn-in for each sampler; the pCN cross-check runs twice as long per chain, importance sampling draws 6000 proposals from the Laplace Gaussian, and the coverage sweep refits $R = 24$ noise replications for each of six truth/noise configurations, scoring a 1,363-node evaluation grid on each replication for 32,712 coverage tests per configuration.

The exact Dupire operator. The weighted layer (5) is implemented in `src/ext/common.py` as a separate differentiable operator in which the stiffness $\mathbf{K}(w)$ with $w = e^{-k}$ is independent of the learned coefficient and only the weighted mass $\mathbf{M}(m_T)$ with $m_T = e^{-k}/\kappa$ depends on it. The implementation is verified against the Black–76 closed form for a constant coefficient, confirming the derivation of (4) and the SPD structure. E1, E2, E6 and E7 execute this operator and store its provenance in their JSON sidecars. E3–E5 and E8–E11 intentionally execute the symmetric benchmark because their stated targets are regularisation, obstacle geometry, constraint placement, operator shift, score estimation and UQ under that operator; their numbers are not relabelled as exact-Dupire results.

B CBOE data pipeline

OCC symbols are parsed as `root + YYMMDD + C/P + strike` $\times 10^3$. We take mid prices $(bid + ask)/2$, drop crossed/zero quotes, recover per-expiry forwards and discount factors by regressing $C - P$ on strike (put–call parity), convert to log-moneyness $k = \log(K/F)$, select OTM options, and compute time-to-maturity on an ACT/365 basis. Expiries with fewer than six clean quotes are discarded.

C Interfaces to current SciML and quantitative-finance frontiers

This section separates mathematical interfaces from empirical contributions. Four of the interfaces below are now backed by controlled experiments: the hard-versus-soft constraint thesis by E8 (10.7), amortisation by E9 (10.8), the diffusion/score connection by E10 (10.9), and Laplace uncertainty quantification by E11 (10.10). Two remain purely mathematical specifications: the entropic martingale bridge (C.4) and the rough/nonlocal extensions (C.5). We keep that distinction explicit throughout.

C.1 From soft physics penalties to a hard-constrained neural PDE solver

A PINN represents the state by a network u_φ and minimises collocation or variational residuals. Deep Ritz methods minimise an energy over a neural trial space. Our architecture makes a different approximation choice:

$$f_\theta \mapsto \kappa_\theta \mapsto \{\mathbf{M}, \mathbf{K}\} \mapsto u_h,$$

where only the coefficient is learned and u_h is the solution of the assembled Galerkin equations. Positivity of κ_θ , coercivity of the form, and SPD of the implicit layer hold independently of the loss weight and training accuracy. This is analogous in spirit to Hamiltonian, Lagrangian and port-Hamiltonian networks, which place invariants or dissipation in the model class, but the preserved object here is an elliptic energy and its discrete linear algebra rather than a symplectic flow. 3 makes the separation sharp: for a soft formulation the inadmissible set is never empty at any finite penalty weight, whereas the reparameterisation (9) makes the whole parameter space admissible. A meaningful PINN/Deep-Ritz comparison must therefore measure not only price error but also failed factorisations, per-layer spectra, gradient norms, coefficient admissibility, and transferability of the recovered coefficient. E8 (10.7) runs exactly that comparison and finds the predicted behaviour: all three soft baselines fit their own prices to within a few percent while driving the coefficient negative on 6.9%–55.4% of the grid. The five-seed formal aggregate has mean PINN Cholesky-failure count 0.0, while the separate seed-11 profile exhibits 5 indefinite layers and the corresponding failed

factorisations. 1 explains the remaining two profile cases, whose lumped mass masks the violation at the linear-algebra level but cannot prevent the resulting loss of transferability.

C.2 Amortisation as a structure-preserving neural operator

DeepONet and FNO learn a map between function spaces from many paired PDE instances. The present σ -Q map is a differentiable numerical coefficient-to-solution operator, not yet an amortised neural operator. A principled hybrid would learn only a preconditioner or a low-dimensional coefficient code across tasks,

$$\mathcal{G}_\psi : (\text{quote set}, \text{boundary data}) \mapsto f, \quad u = \mathcal{Q}(\text{softplus}(\mathcal{G}_\psi) + \kappa_{\min}), \quad (25)$$

while retaining the Galerkin solve as the final layer.

Formally, the σ -Q map is already an operator between function spaces rather than between grids. Writing $V_h = \text{span}\{\phi_i\} \subset H^1(\Omega)$ for the P_1 space and Π_h for its nodal interpolant, the composite

$$\mathcal{Q}_h : L_+^\infty(\Omega_T) \longrightarrow C([0, T]; V_h), \quad \mathcal{Q}_h = \mathcal{S}_h \circ \mathcal{A}_h, \quad (26)$$

factors into an assembly step $\mathcal{A}_h : \kappa \mapsto \{\mathbf{M}, \mathbf{K}(\Pi_h \kappa)\}$ and a solve step $\mathcal{S}_h$. The distinction from a learned operator is that $\mathcal{Q}_h \rightarrow \mathcal{Q}$ as $h, \Delta\tau \rightarrow 0$ by standard Galerkin convergence, whereas discretisation invariance of an FNO is a statement about *representation* at arbitrary resolution and carries no guarantee that accuracy transfers to a resolution absent from training. E9 (10.8) tests precisely this difference and gives a deliberately mixed verdict. In distribution the FNO is more accurate than the solver (1.93×10^{-3} versus 4.36×10^{-3} relative L^2), because it learns to correct the coarse-grid discretisation error that a consistent scheme commits; under strike refinement the ordering inverts (2.33×10^{-3} versus 3.53×10^{-3}); and under horizon extrapolation the learned operator’s error is approximately 5.9× the solver error (3.59×10^{-2} versus 6.12×10^{-3}) while emitting negative prices on 21.6% of nodes. Projecting the learned output onto the admissible box afterwards removes box violations by construction but still leaves 10.7% monotonicity violations and does not recover accuracy. The relevant comparison is thus not “solver versus FNO” but black-box amortisation versus amortisation whose final layer is a structure-preserving solve; E9 supports the latter under distribution shift and is honest about the in-distribution cost.

C.3 Forward diffusion, score models, and a learned noise geometry

6 established that the density dual (19) is the Fokker–Planck equation of the log-forward diffusion (20), and 7 gave its exact reverse-time drift (22). This is the precise connection to score-based generative modelling: both are governed by a forward Fokker–Planck equation and a reverse-time dynamics whose drift depends on a density score [20, 21]. There are also crucial differences. In a conventional diffusion model the noising schedule is chosen to map data to a simple reference distribution; here the state-dependent diffusion field is inferred from option prices and must respect financial marginals. It is therefore best interpreted as a *market-calibrated anisotropic noise geometry*, not as a DDPM schedule in disguise, and 3 shows the concrete consequence: the calibrated field contributes a transport term $2\partial_k \kappa$ that a constant-variance schedule does not have.

The Galerkin construction also yields a generative-modelling asset rather than merely an analogy. Because $q_T = \partial_k(e^{-k}\partial_k c)$ is produced by the same SPD solve as the price, the score is available by post-processing with no training whatsoever. E10 (10.9) quantifies this. The discrete mass defect $|\int q - 1|$ behaves exactly as the boundary bracket in the proof of 6(ii) predicts, falling from 4.2×10^{-4} to 1.6×10^{-6} under refinement in k while remaining flat at 4.1×10^{-7} when only the

time step is refined. Sampling (20) reproduces the operator density to within the Monte-Carlo floor at every step count tried (W_1 between 4.7×10^{-4} and 6.2×10^{-4} , against a self-distance floor of $5.7 \times 10^{-4} \pm 1.5 \times 10^{-4}$). The training-free score attains relative L^2 error 2.06×10^{-2} , against 9.5×10^{-2} for converged denoising score matching and 0.92 for implicit score matching at the same sample size; fed into (22) it samples at $0.74\times$ the Monte-Carlo floor, versus $6.2\times$ for the best score-matching model and $1.67\times$ for a tuned variance-preserving DDPM, at 6.9 ms instead of tens of seconds of training. The scope of that claim is narrow by design: the benchmark is a one-dimensional density with a closed-form score, so it demonstrates that the dual view is a genuine mass-conservative score model and that SPD assembly is enough to obtain a competitive score, not that Galerkin scores are preferable in high-dimensional generative modelling.

C.4 Martingale optimal transport and Schrödinger bridges

Vanilla prices identify marginal laws but not the joint path law. Martingale optimal transport (MOT) studies the family of martingale couplings compatible with those marginals and yields robust bounds for path-dependent claims [22]. A Schrödinger bridge instead selects a path measure by minimising relative entropy to a reference diffusion subject to marginal constraints [23]. These are complementary to, rather than rephrasings of, local-volatility inversion.

A disciplined integration would use the differentiable SPD solver as an inner map in an outer constrained problem: observed option prices determine marginal penalties, martingale constraints enter through dual variables, and the inner Galerkin layer propagates a candidate diffusion. For a fixed reference diffusion, the classical bridge primarily changes drift; changing diffusion can make path measures mutually singular because quadratic variation changes. Consequently, the Dirichlet energy in (7) cannot simply be called a Schrödinger entropy. An entropic martingale bridge would be a new model and would require its own feasibility, duality, and multi-marginal experiments.

C.5 Rough volatility and positive nonlocal forms

Rough volatility is predominantly a non-Markovian *temporal-memory* phenomenon, whereas a spatial fractional Laplacian describes jumps or nonlocal state propagation; the two should not be conflated. A rough extension of (10) would replace the local time derivative or Markov state by a Volterra history operator, as in rough Bergomi or rough Heston models [24, 25, 26]. A spatial nonlocal extension would instead use a symmetric kernel form

$$a_J(u, v) = \frac{1}{2} \int_{\Omega} \int_{\Omega} J_{\theta}(k, k') (u(k) - u(k')) (v(k) - v(k')) dk dk', \quad (27)$$

with $J_{\theta}(k, k') = J_{\theta}(k', k) \geq 0$. This condition produces a positive semidefinite dense or low-rank stiffness matrix and is the genuine nonlocal analogue of the present hard constraint. Asymmetric Lévy kernels generally lose symmetry. Either route changes the model class and belongs to a separate study; local-volatility marginals alone cannot identify rough path dependence.

C.6 Adjoint-enabled uncertainty quantification

The current estimator is deterministic, but the existing adjoint makes a local Bayesian approximation comparatively inexpensive. Let f denote the pre-softplus field, $y = Q(\kappa(f)) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \Gamma)$, and place a proper Gaussian prior $f \sim \mathcal{N}(m_0, C_0)$. At the MAP estimate $\hat{f}$, the Gauss–Newton Laplace approximation is

$$f | y \approx \mathcal{N}(\hat{f}, H_{\text{GN}}^{-1}), \quad H_{\text{GN}} = J^{\top} \Gamma^{-1} J + C_0^{-1}, \quad (28)$$

where $J = D_f \mathcal{Q}(\kappa(f))|_{\hat{f}}$ is obtained from the tangent/adjoint solves in 5. Pointwise bands for any smooth functional $g(f)$ — in particular the local-volatility surface $\sigma_{\text{loc}} = \sqrt{2\kappa(f)}$ and the dual density row $q_T = \partial_k(e^{-k}\partial_k c)$ — follow by the delta method,

$$\text{Var}(g(f)) \approx (D_f g) H_{\text{GN}}^{-1} (D_f g)^\top, \quad (29)$$

with $D_f g$ assembled from the same tangent solves. The computation never requires forming H_{GN} densely: Hessian-vector products with conjugate gradients suffice in the large-parameter regime, and we form the dense factor only because our Chebyshev parameterisation is 36-dimensional.

A recurring source of confusion in inverse-problem UQ is that the Laplace posterior spread and the sampling spread of the penalised point estimator are not estimates of the same quantity, and the two are ordered. The next statement makes the ordering exact, which is what allows us to read a systematically *narrower* bootstrap spread as a consistency check rather than as evidence against the Laplace approximation.

Proposition 8 (Sandwich domination of the penalised estimator). *Let $\hat{f}$ solve the penalised least-squares problem $\min_f \frac{1}{2}\|y - \mathcal{Q}(\kappa(f))\|_{\Gamma^{-1}}^2 + \frac{1}{2}\|f - m_0\|_{C_0^{-1}}^2$ with $C_0 \succ 0$, write $J = D_f \mathcal{Q}(\kappa(f))|_{\hat{f}}$, $F = J^\top \Gamma^{-1} J$ and $H = F + C_0^{-1}$, and let $\Sigma = H^{-1}$ be the Gauss–Newton Laplace covariance (28). In the linearised regime the estimator satisfies $\hat{f} - f_\star \approx H^{-1} J^\top \Gamma^{-1} \varepsilon$ up to the deterministic prior pull, so its sampling covariance is the sandwich form*

$$S = H^{-1} F H^{-1} = \Sigma - H^{-1} C_0^{-1} H^{-1} \preceq \Sigma, \quad (30)$$

with equality if and only if $C_0^{-1} = 0$. Consequently every marginal standard deviation of the penalised estimator is bounded above by the corresponding Laplace posterior standard deviation, and the gap is largest in the directions the prior actually constrains.

Proof. Differentiating the first-order condition $-J^\top \Gamma^{-1} (y - \mathcal{Q}(\kappa(\hat{f}))) + C_0^{-1} (\hat{f} - m_0) = 0$ and linearising $\mathcal{Q}(\kappa(f)) \approx \mathcal{Q}(\kappa(f_\star)) + J(f - f_\star)$ with $y = \mathcal{Q}(\kappa(f_\star)) + \varepsilon$ gives $H(\hat{f} - f_\star) = J^\top \Gamma^{-1} \varepsilon - C_0^{-1} (f_\star - m_0)$. The second term is deterministic, hence $\text{Cov}(\hat{f}) = H^{-1} J^\top \Gamma^{-1} \text{Cov}(\varepsilon) \Gamma^{-1} J H^{-1} = H^{-1} F H^{-1} = S$ using $\text{Cov}(\varepsilon) = \Gamma$. For the identity in (30) substitute $F = H - C_0^{-1}$: $S = H^{-1} (H - C_0^{-1}) H^{-1} = H^{-1} - H^{-1} C_0^{-1} H^{-1} = \Sigma - H^{-1} C_0^{-1} H^{-1}$. Since $C_0 \succ 0$ and $H \succ 0$, the matrix $H^{-1} C_0^{-1} H^{-1}$ is symmetric positive definite, so $\Sigma - S \succ 0$ and $S \prec \Sigma$; taking the i -th diagonal entry of $\Sigma - S$ gives $S_{ii} < \Sigma_{ii}$ for every i , i.e. marginal domination. Equality in (30) requires $H^{-1} C_0^{-1} H^{-1} = 0$, i.e. $C_0^{-1} = 0$. Finally, writing the gap in the eigenbasis of the prior shows that its size in direction v is $\|C_0^{-1/2} H^{-1} v\|^2$, which vanishes for directions the data determine ($F \gg C_0^{-1}$, so $H^{-1} \approx F^{-1}$ annihilates the prior scale) and is maximal for prior-dominated directions. $\square$

Remark 4 (What each interval answers). 8 implies that a parametric bootstrap of the *penalised* estimator must, in the linear regime, report smaller spread than the Laplace posterior — the bootstrap resamples noise around a fixed regulariser and therefore inherits the shrinkage, whereas the posterior also carries the prior’s own uncertainty. A bootstrap-to-Laplace standard-deviation ratio below one is thus the predicted behaviour, and the informative check is whether the bootstrap matches the *sandwich* prediction S rather than whether it matches Σ . Validating the Gaussian shape of the posterior itself requires a sampler that targets the same posterior, which is why 10.10 runs both comparisons.

Three cautions prevent “UQ for free” from becoming an overclaim. An H^2 seminorm has a null space and must be completed to a proper prior; weakly identified directions can produce

large posterior variance even when the forward solve is SPD; and the Gaussian approximation can fail near boundaries or active-set changes. 10.10 therefore executes the corresponding study: it reports pointwise coverage of the true surface, compares the Laplace bands against a parametric bootstrap, against the sandwich prediction (30), and against two Metropolis samplers plus self-normalised importance sampling on the same posterior, and quantifies uncertainty amplification in the sparse wings. Bayesian PINNs and modern SciML-UQ benchmarks provide appropriate comparators [27, 28, 29].